

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

Complete

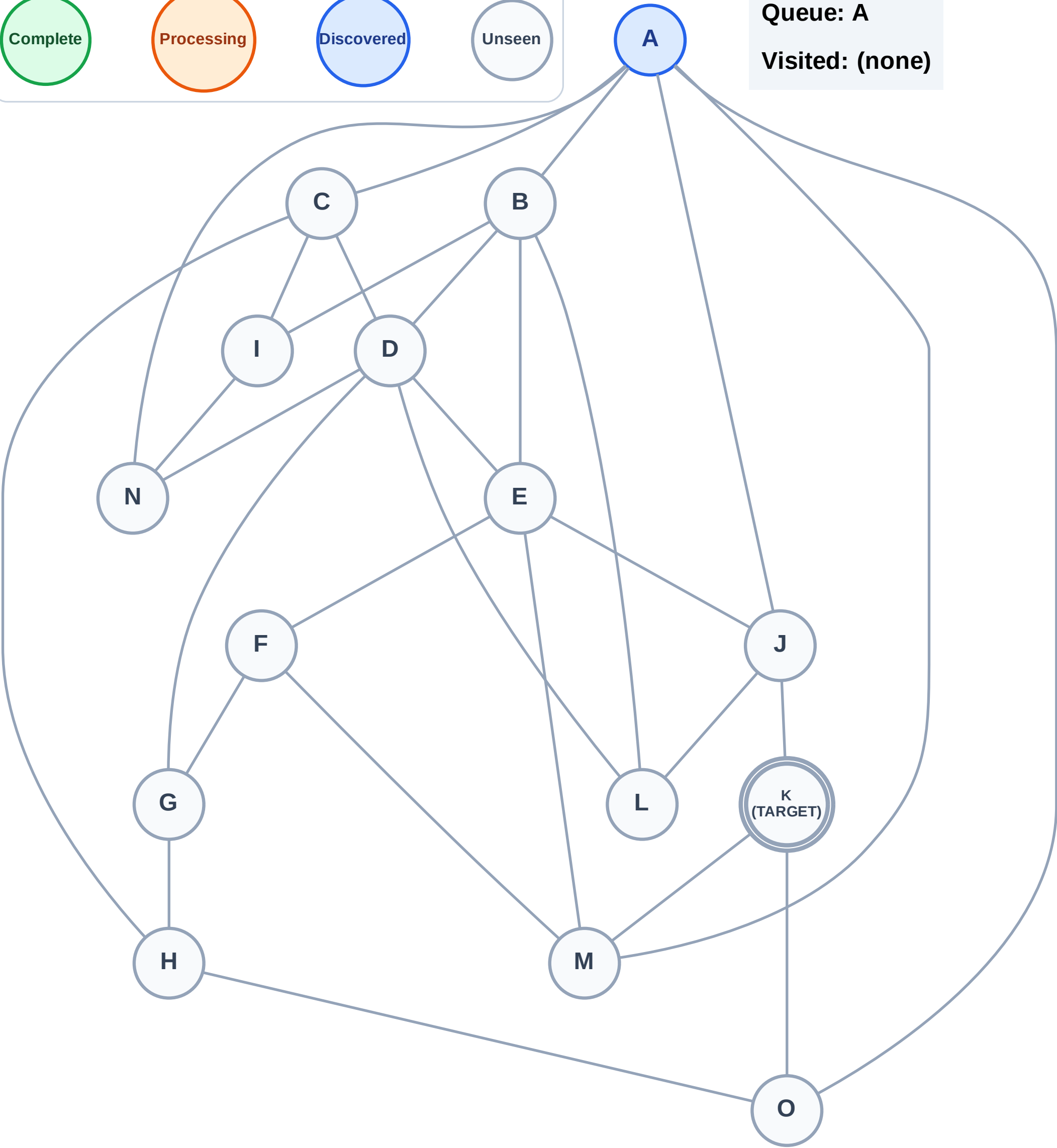
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

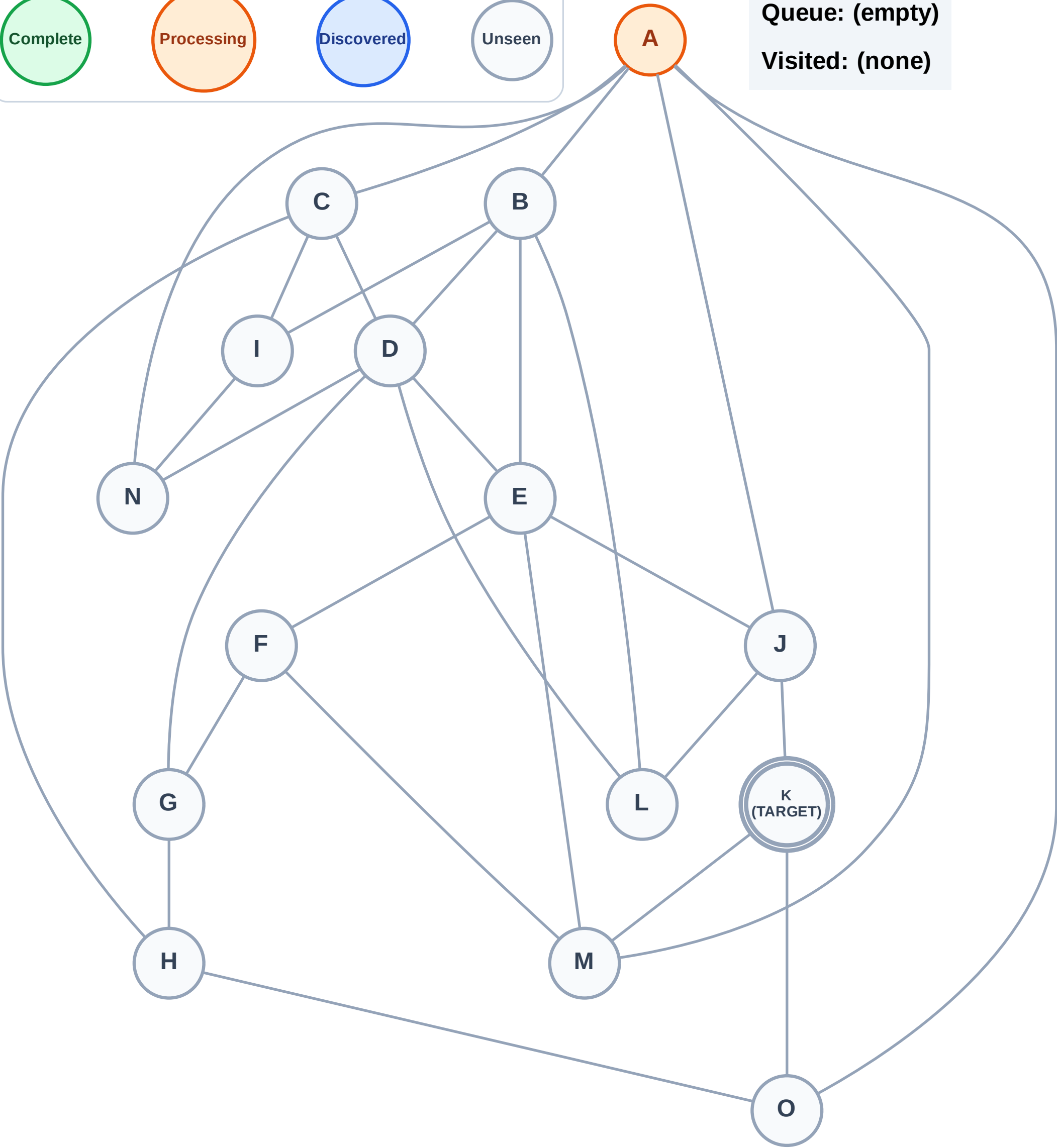
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



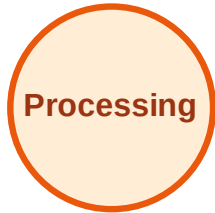
BFS Traversal

Step 3: Discover B, C, J, M, N, O from A

Legend



Complete



Processing



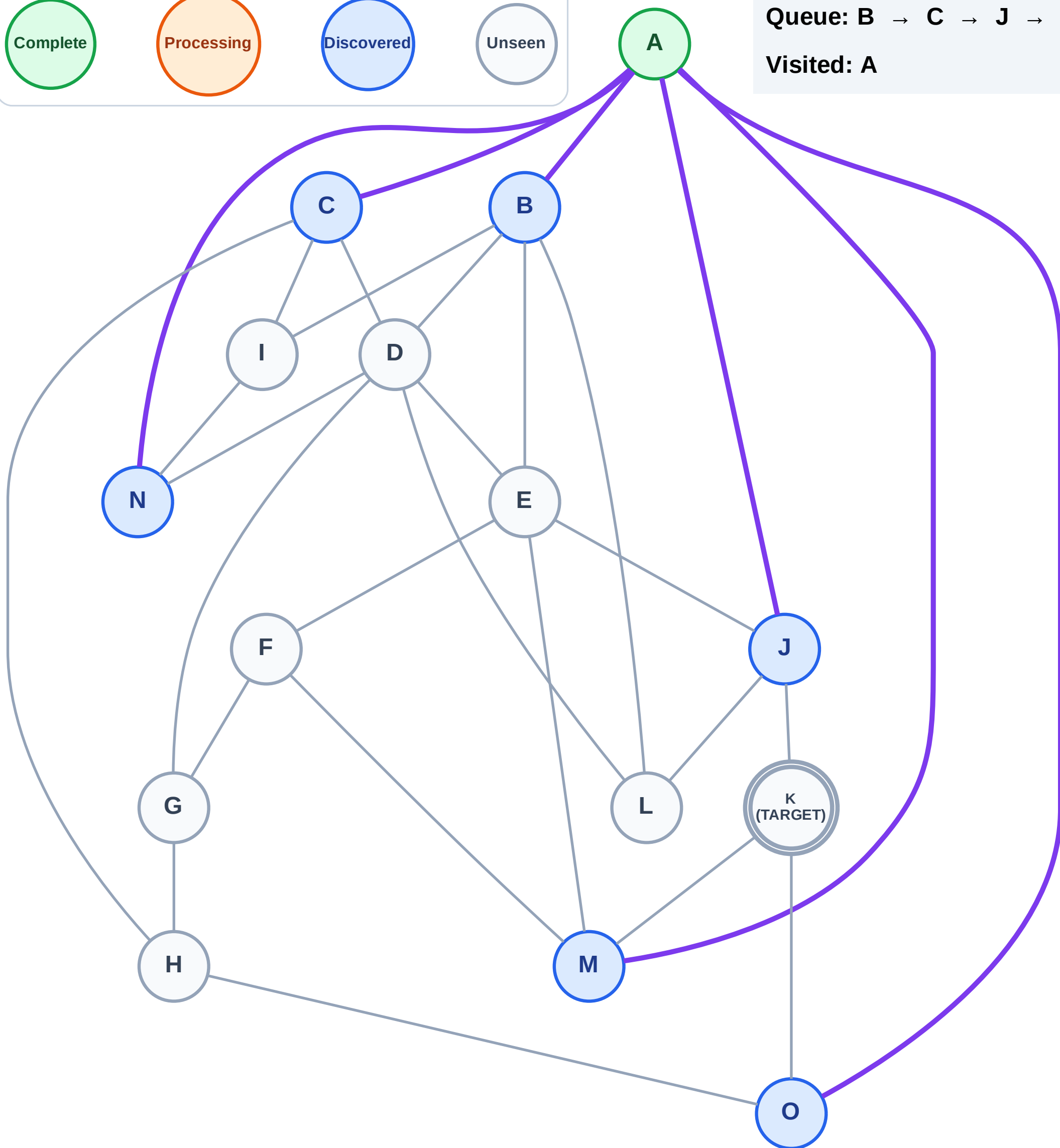
Discovered



Unseen

Queue: B → C → J → M → N → O

Visited: A



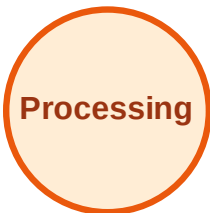
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



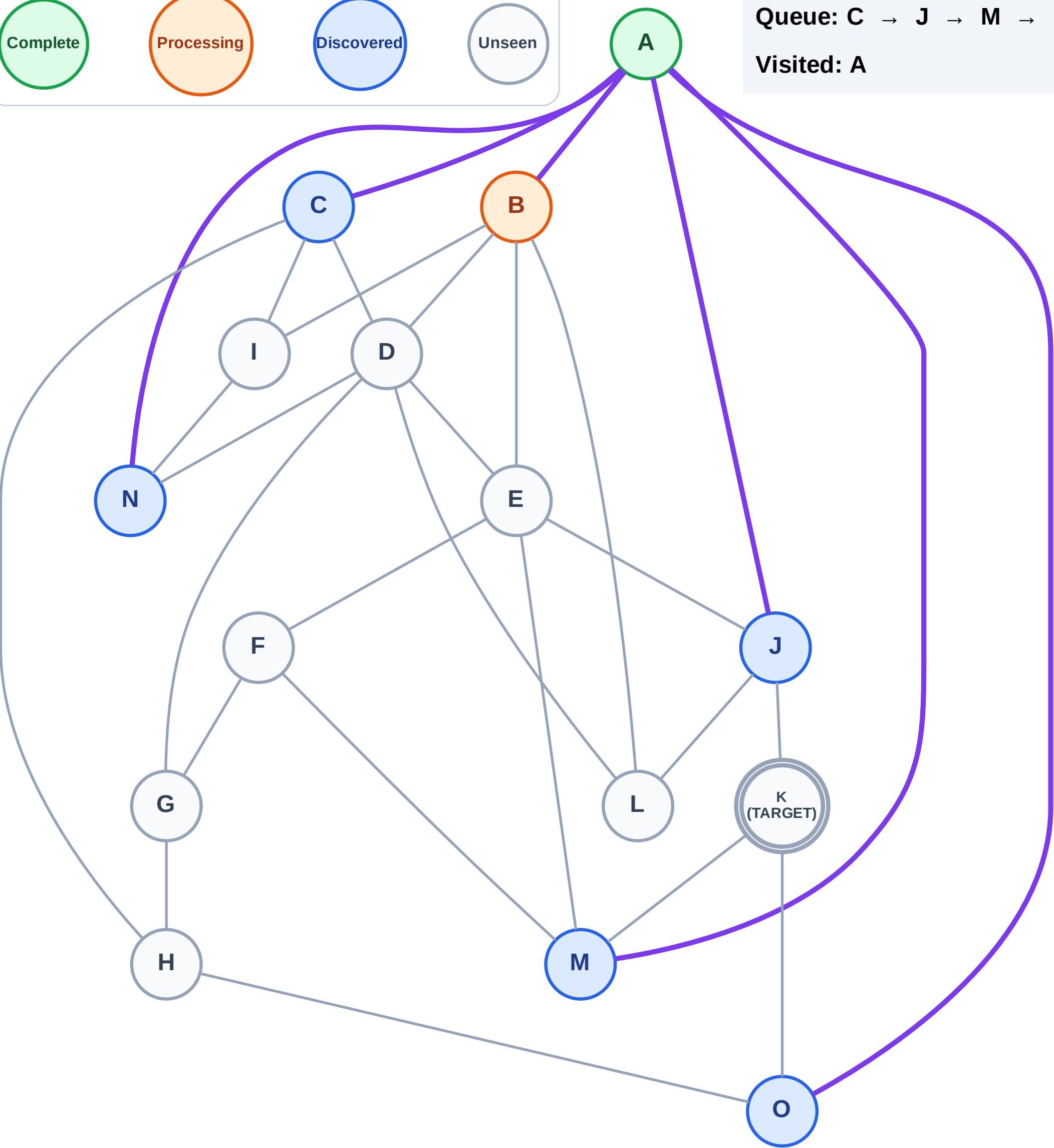
Discovered



Unseen

Queue: C → J → M → N → O

Visited: A



BFS Traversal

Step 5: Discover D, E, I, L from B

Legend

Complete

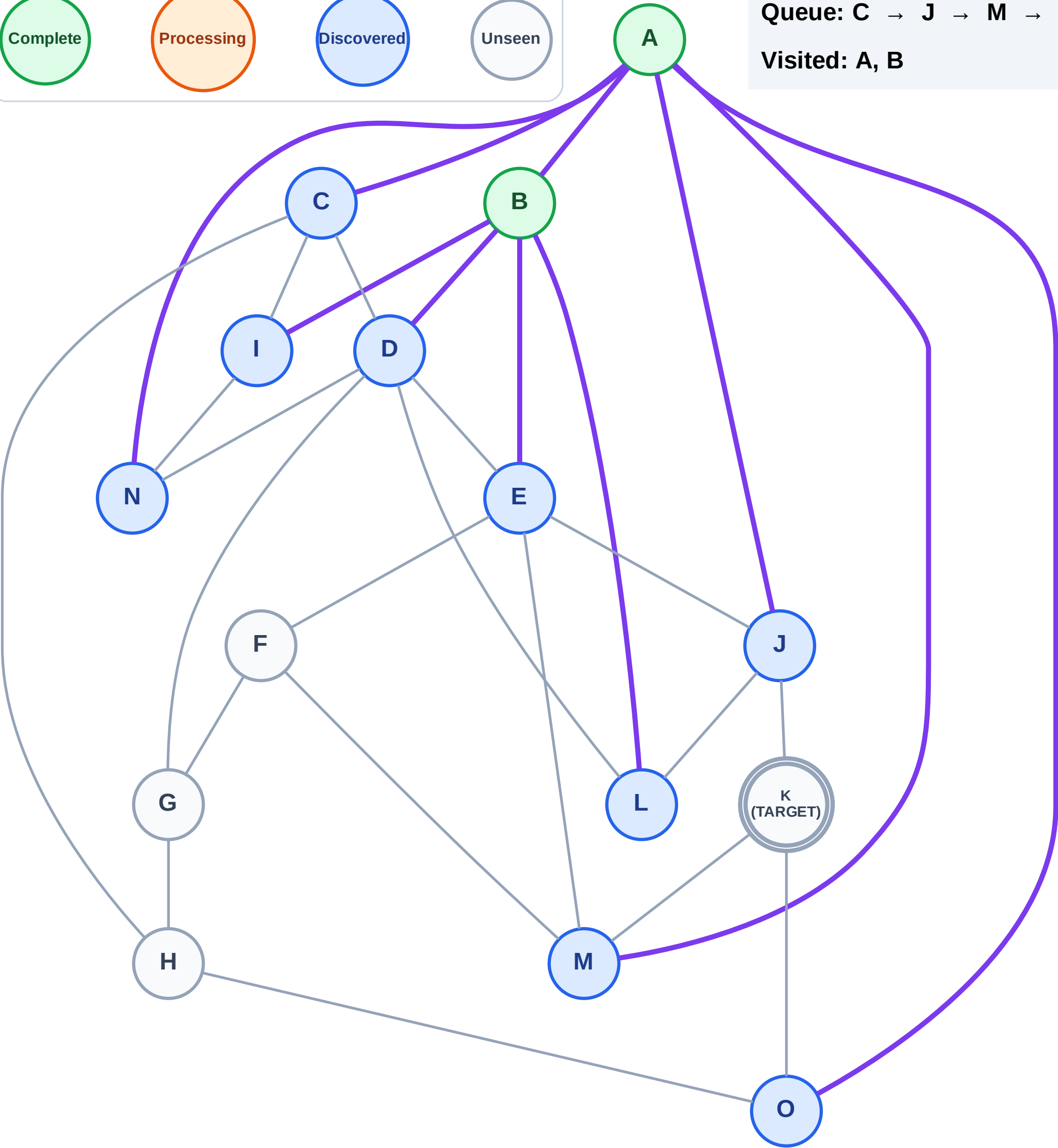
Processing

Discovered

Unseen

Queue: C → J → M → N → O → D → E → I → L

Visited: A, B



BFS Traversal

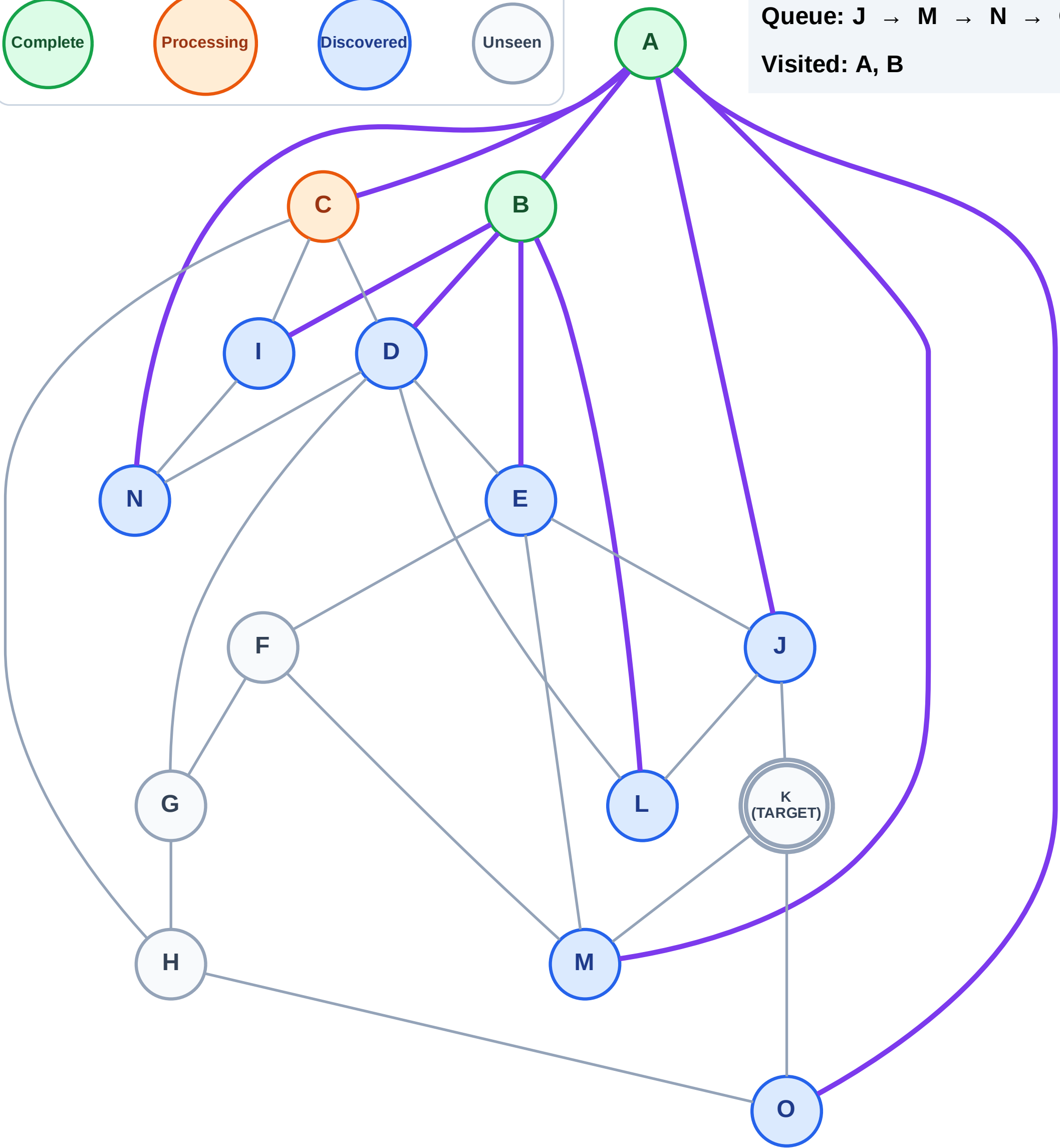
Step 6: Process node C

Legend



Queue: J → M → N → O → D → E → I → L

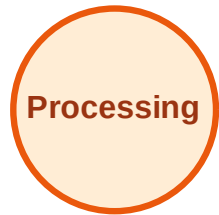
Visited: A, B



BFS Traversal

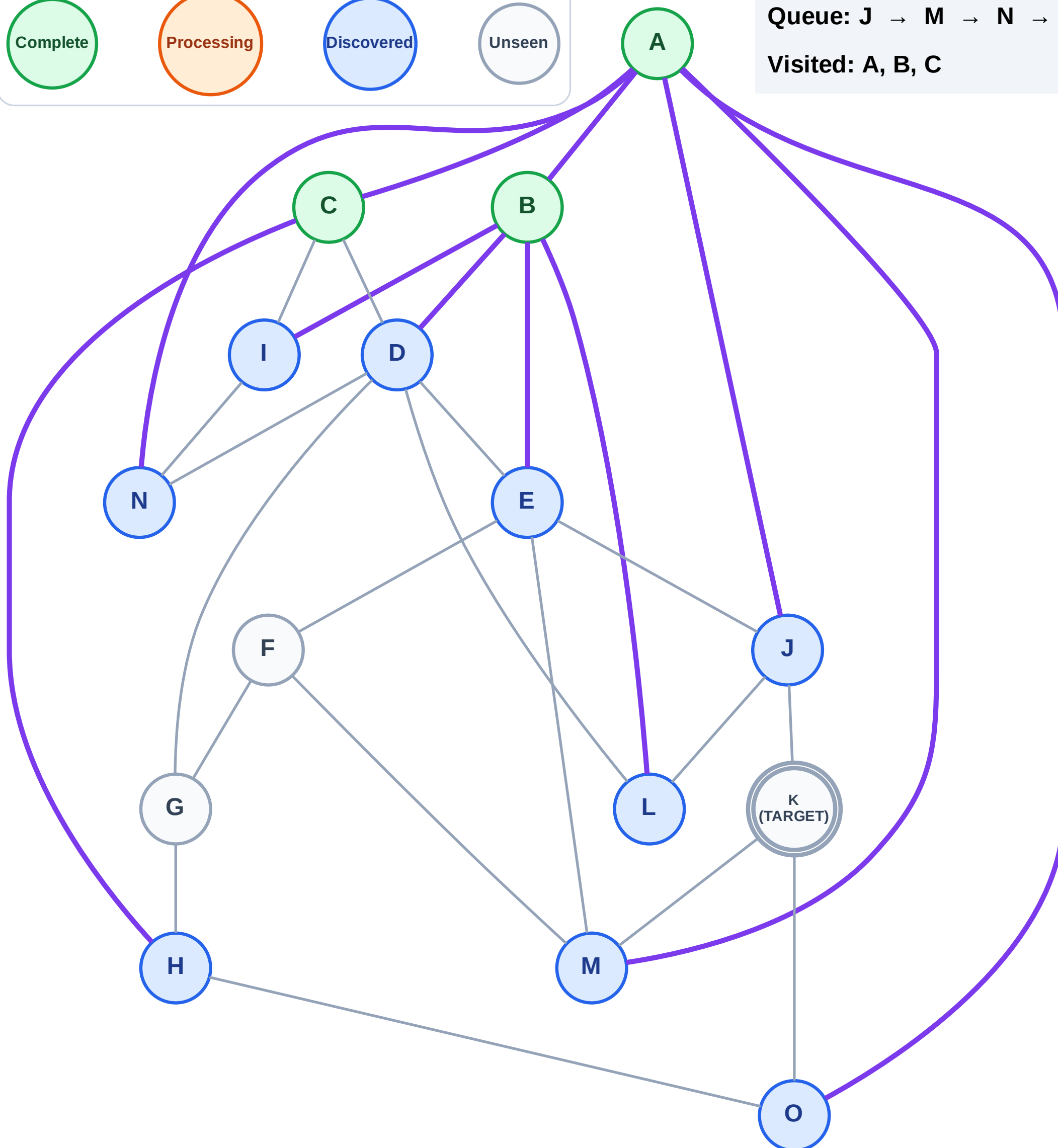
Step 7: Discover H from C

Legend



Queue: J → M → N → O → D → E → I → L → H

Visited: A, B, C



BFS Traversal

Step 8: Process node J

Legend

Complete

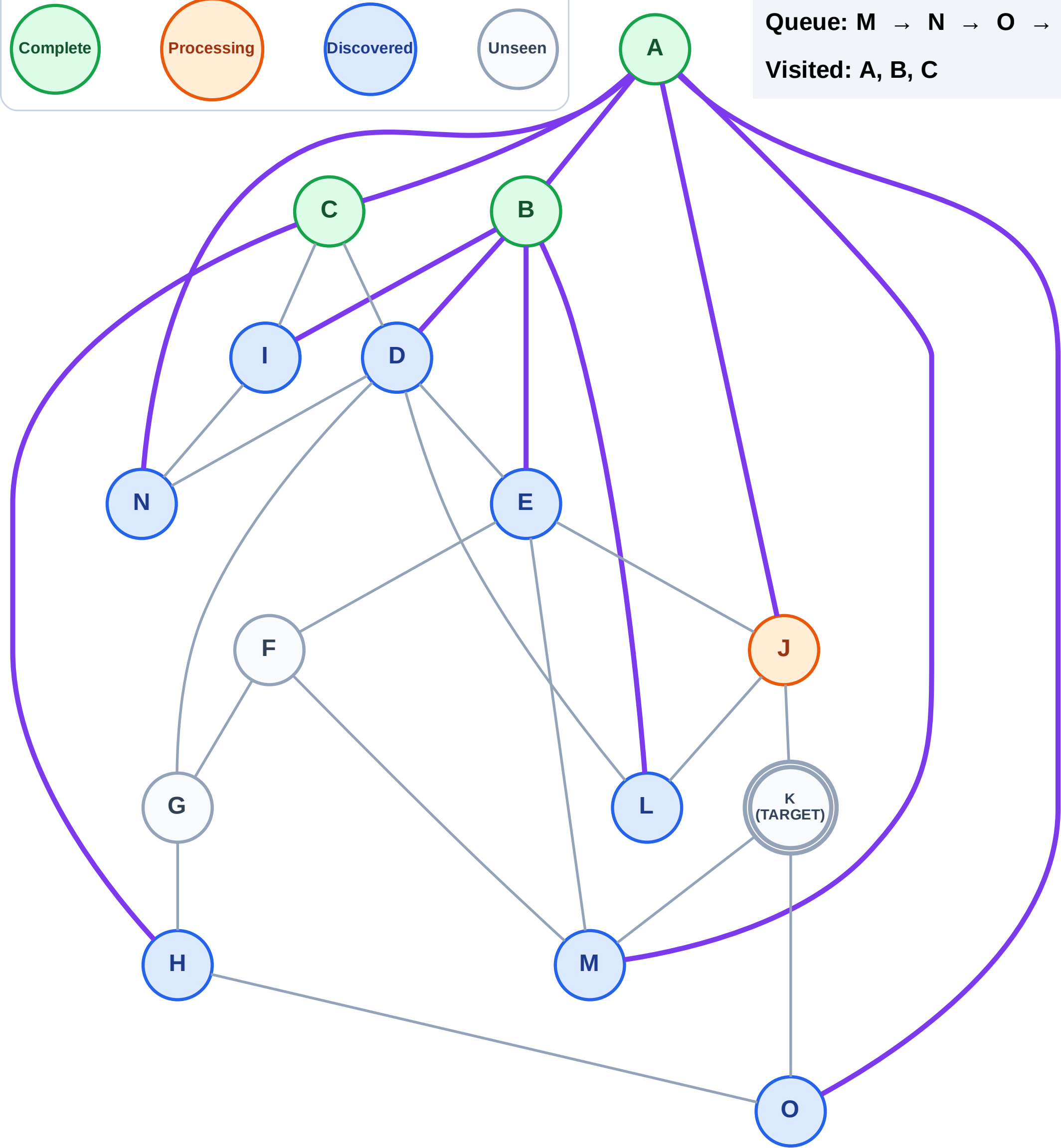
Processing

Discovered

Unseen

Queue: M → N → O → D → E → I → L → H

Visited: A, B, C



BFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → J → K

Legend

Complete

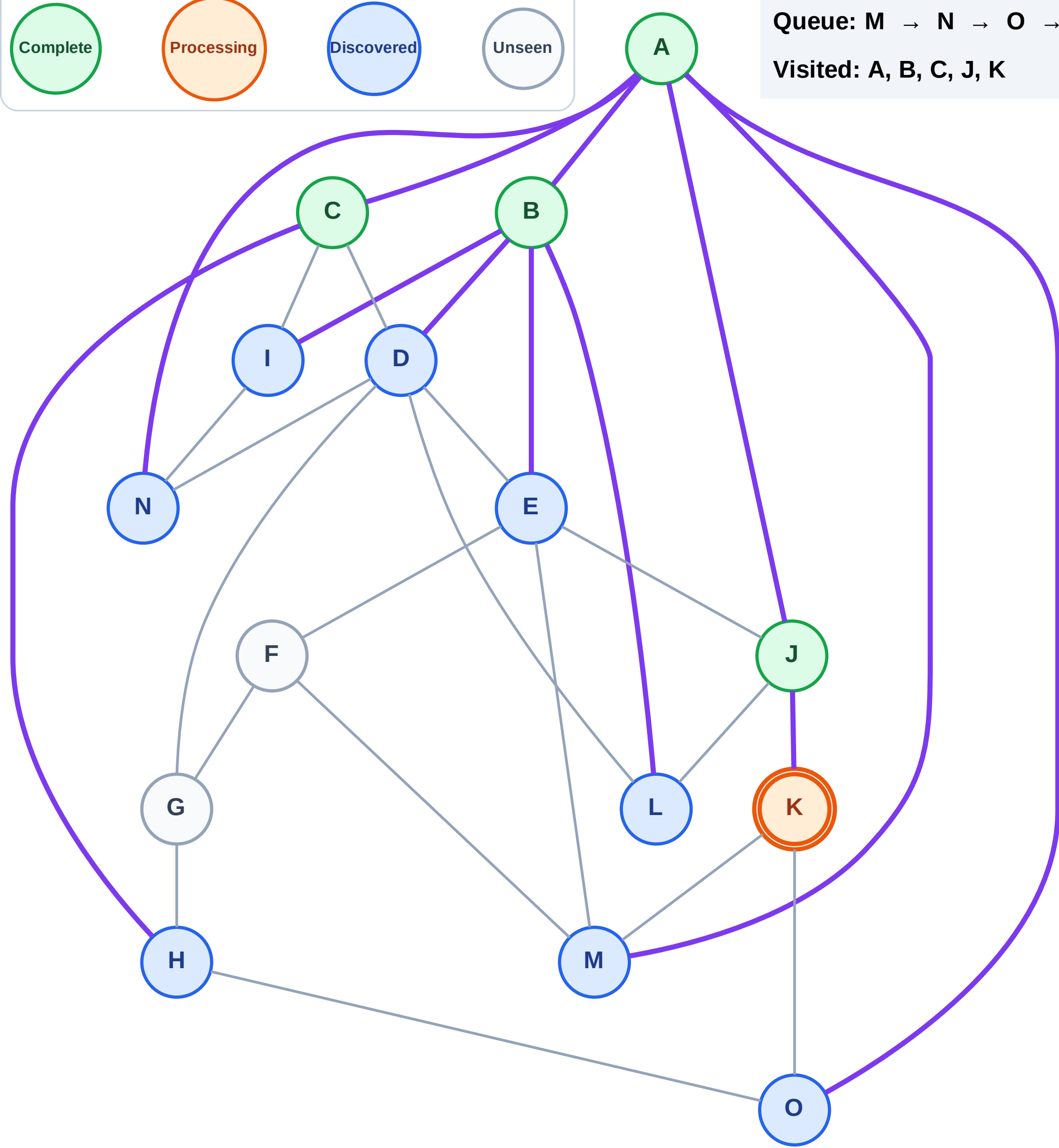
Processing

Discovered

Unseen

Queue: M → N → O → D → E → I → L → H

Visited: A, B, C, J, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

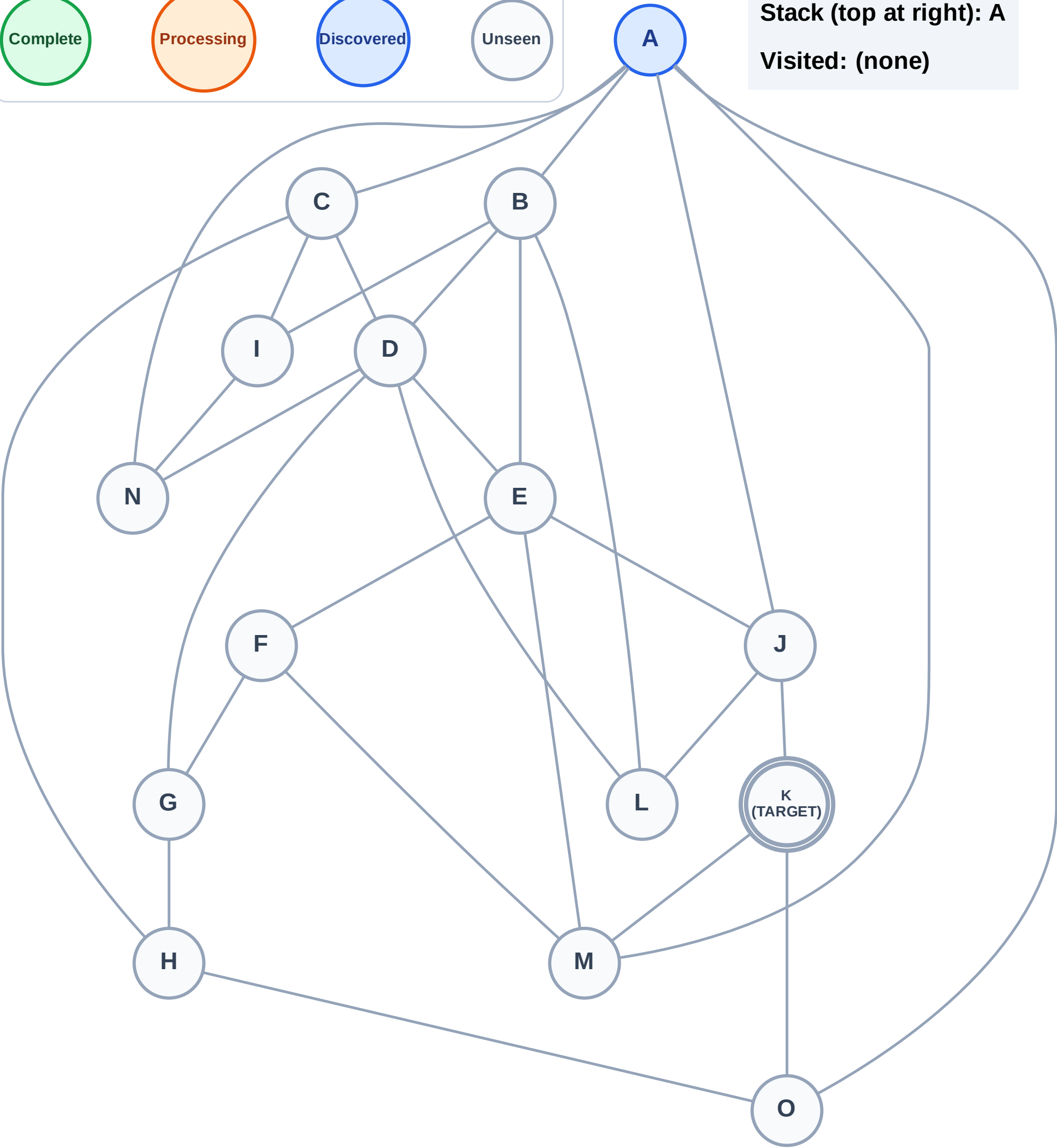
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

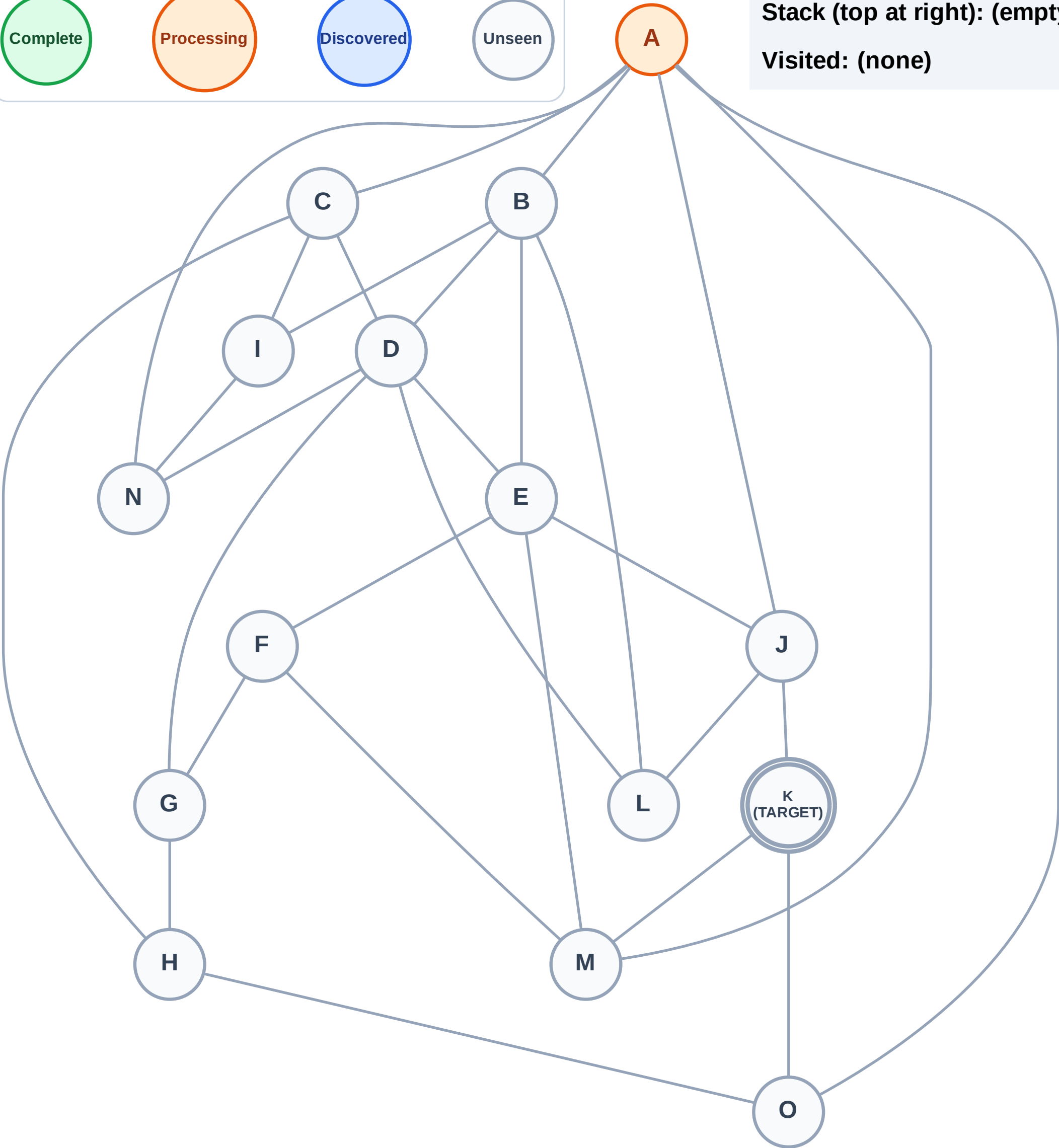
Complete

Processing

Discovered

Unseen

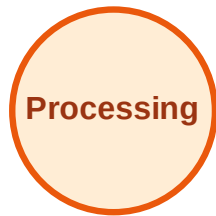
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

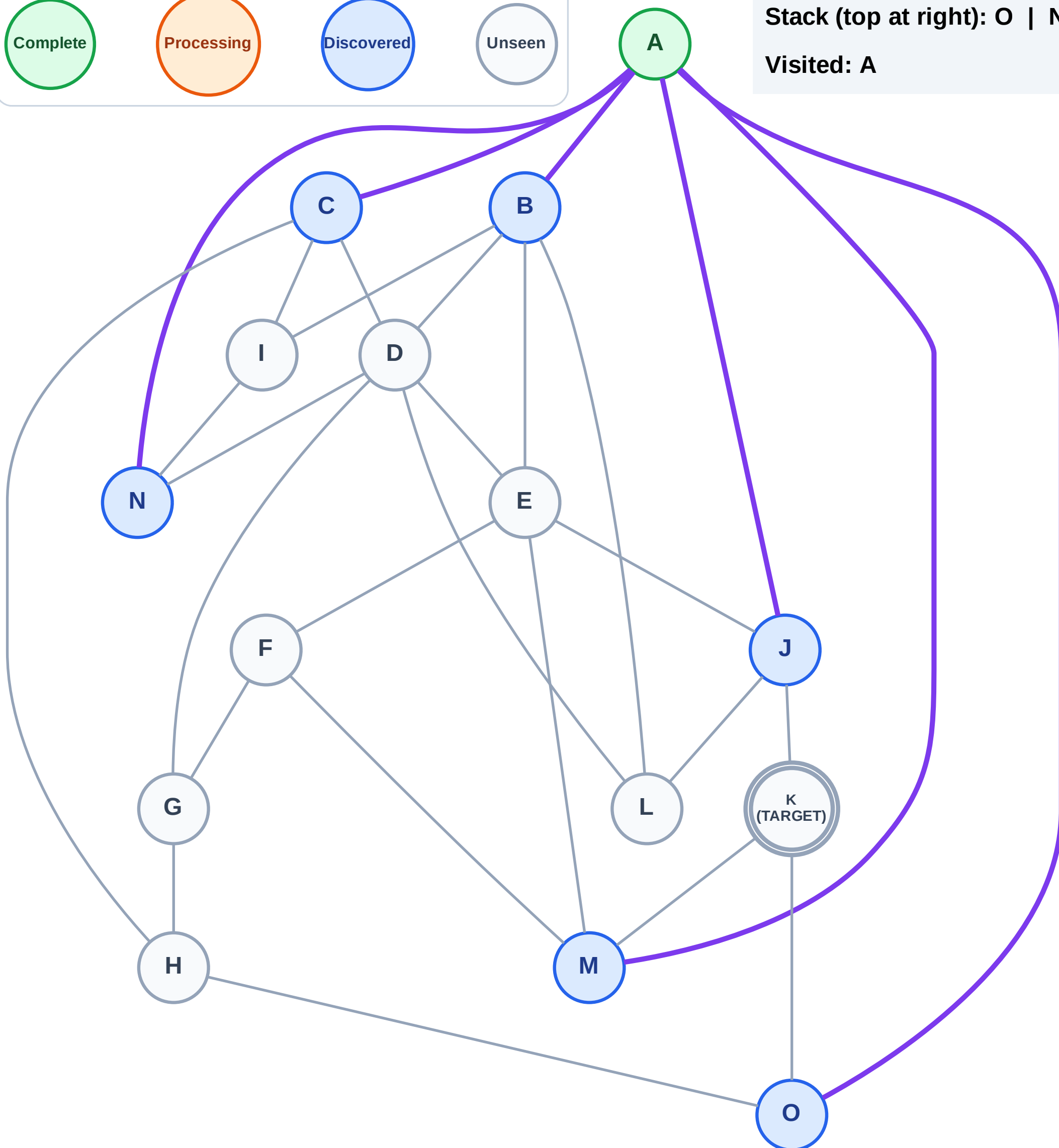
Step 3: Discover O, N, M, J, C, B from A

Legend



Stack (top at right): O | N | M | J | C | B

Visited: A



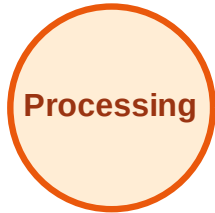
DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



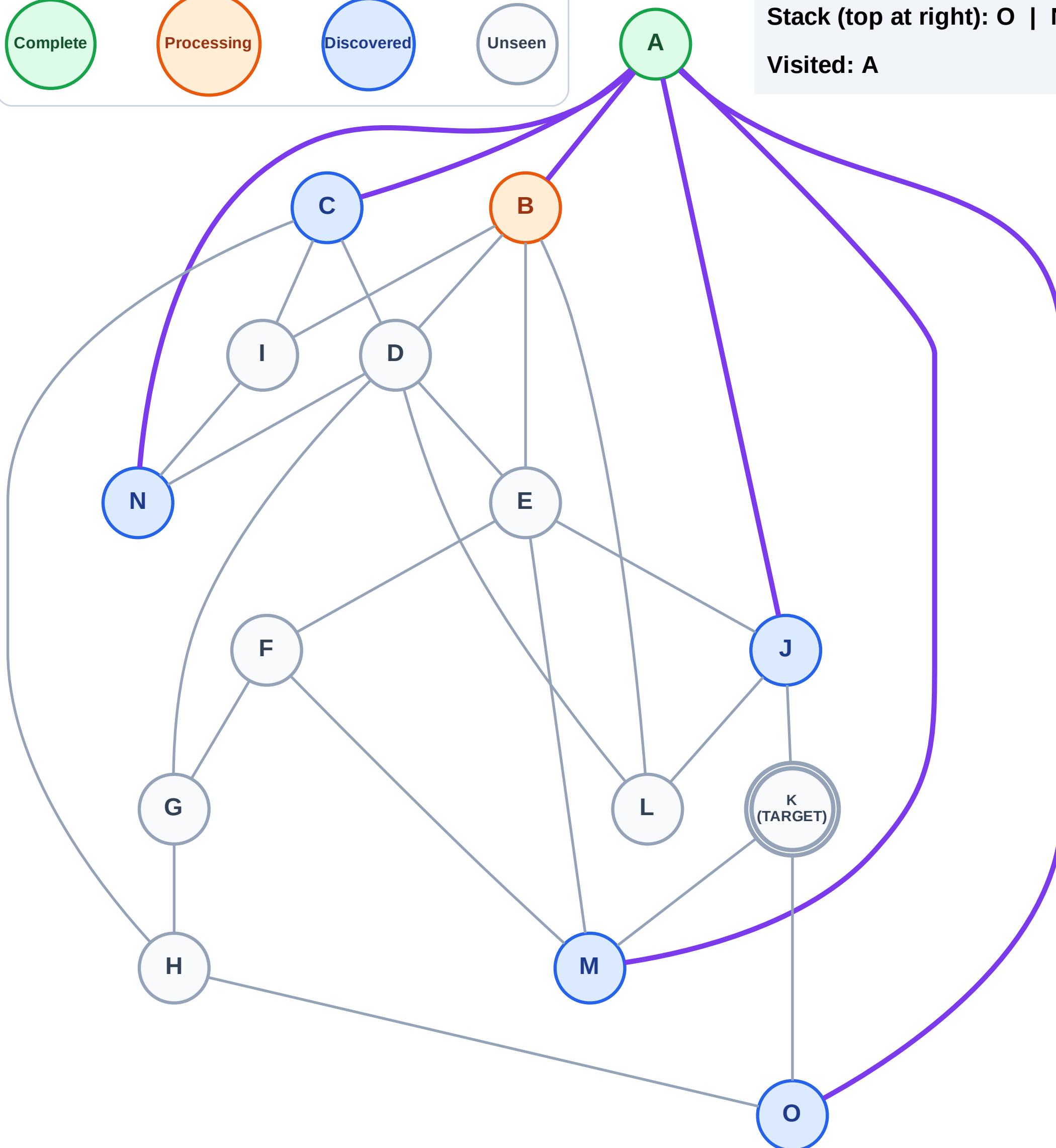
Discovered



Unseen

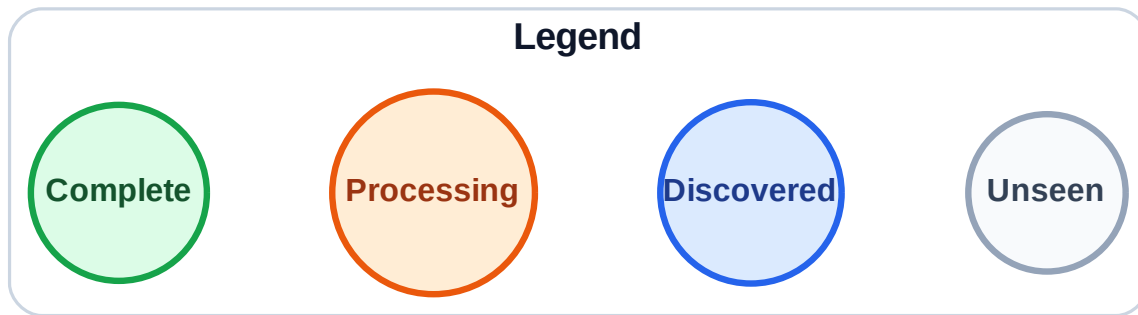
Stack (top at right): O | N | M | J | C

Visited: A



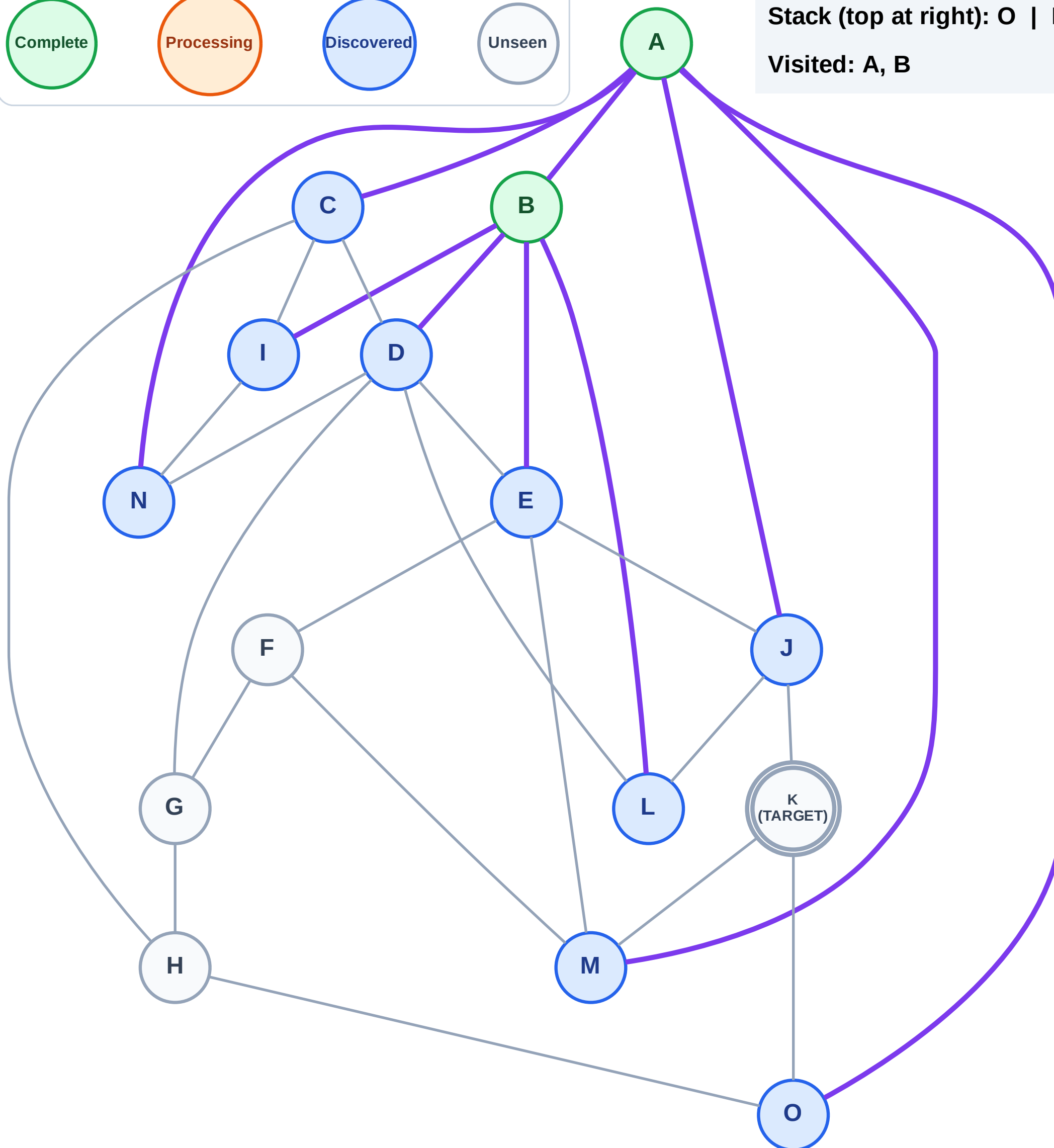
DFS Traversal

Step 5: Discover L, I, E, D from B



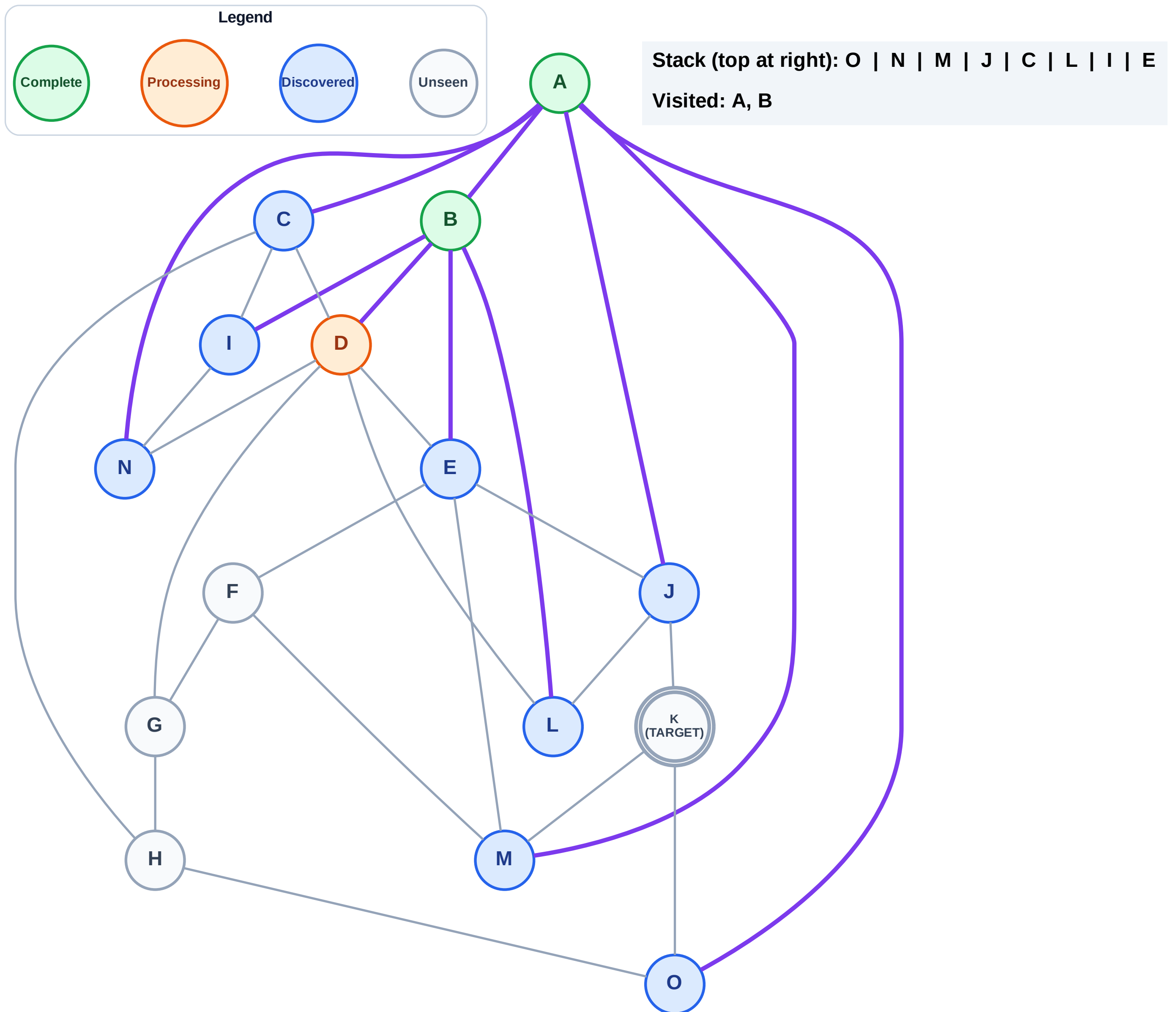
Stack (top at right): O | N | M | J | C | L | I | E | D

Visited: A, B



Step 6: Process node D

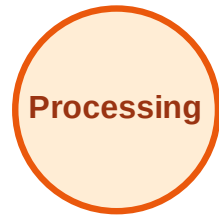
Step 6: Process node D



DFS Traversal

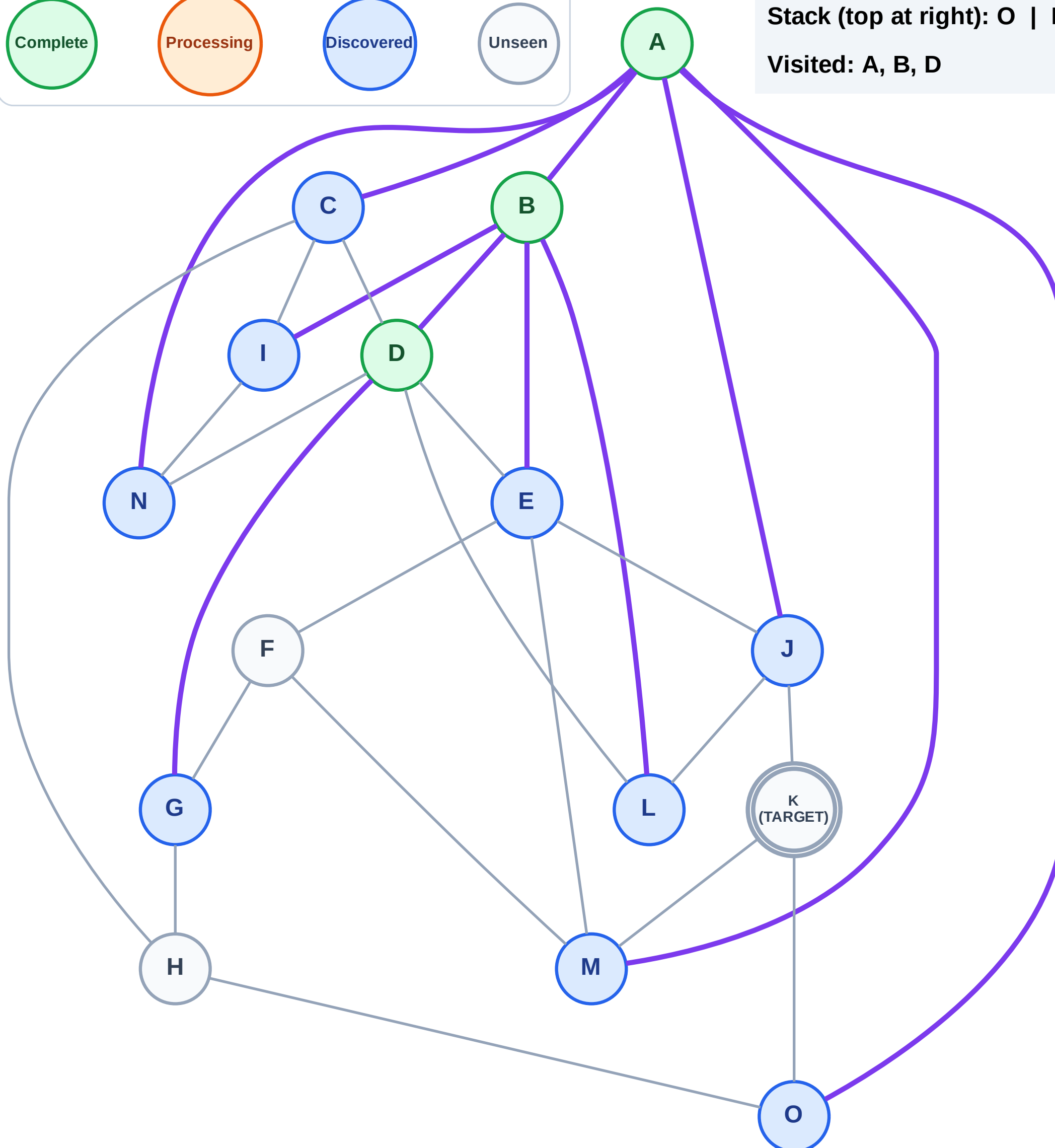
Step 7: Discover G from D

Legend



Stack (top at right): O | N | M | J | C | L | I | E | G

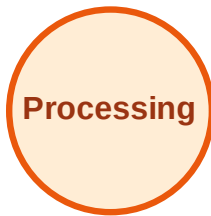
Visited: A, B, D



DFS Traversal

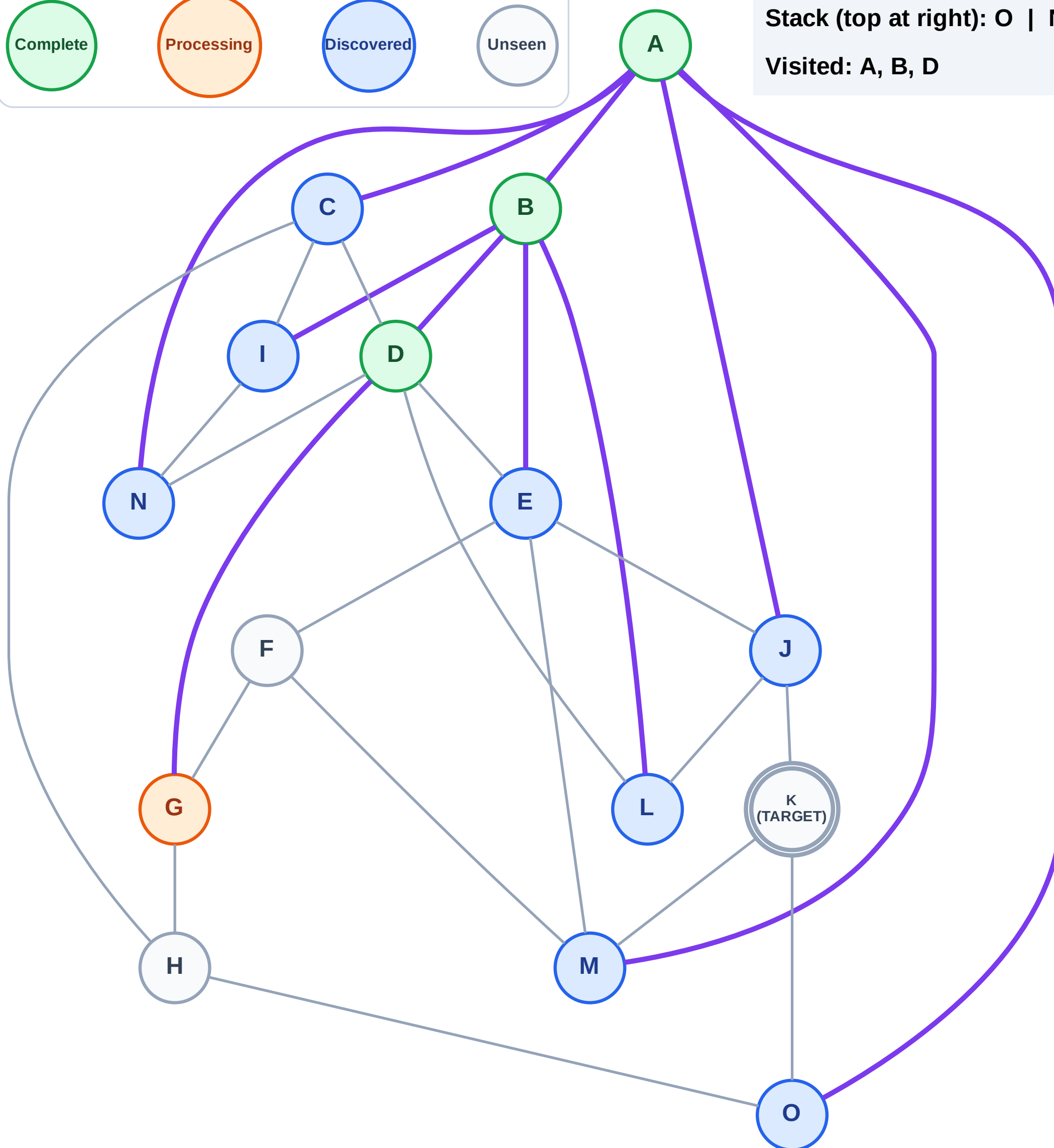
Step 8: Process node G

Legend



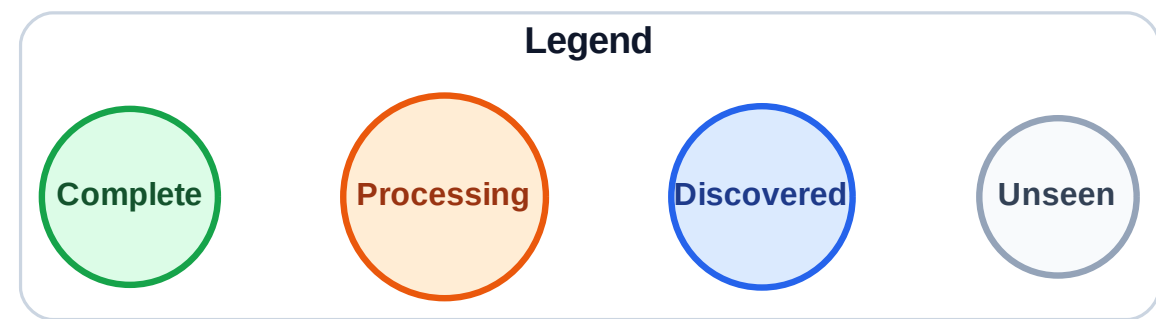
Stack (top at right): O | N | M | J | C | L | I | E

Visited: A, B, D



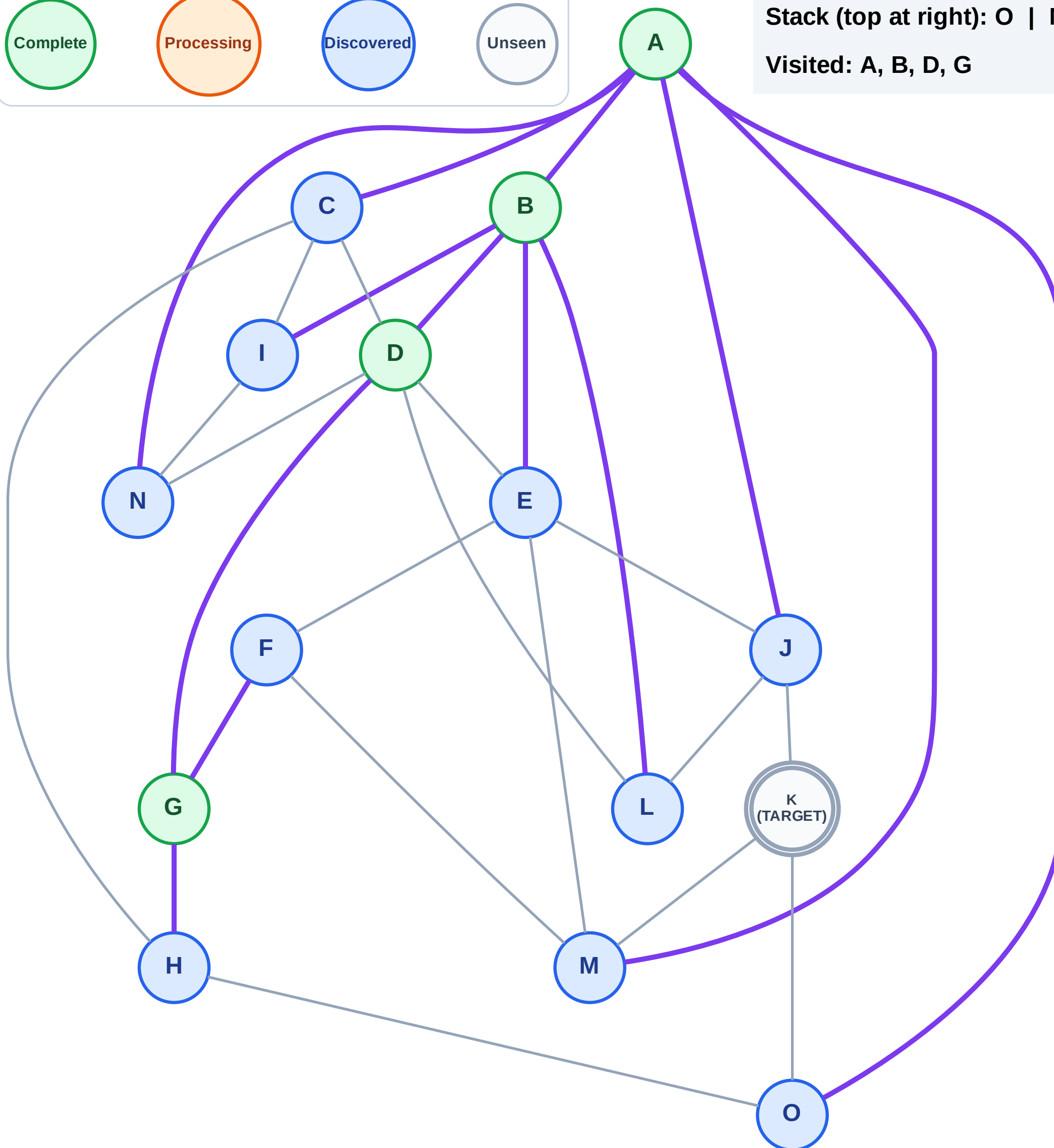
DFS Traversal

Step 9: Discover H, F from G



Stack (top at right): O | N | M | J | C | L | I | E | H | F

Visited: A, B, D, G



DFS Traversal

Step 10: Process node F

Legend

Complete

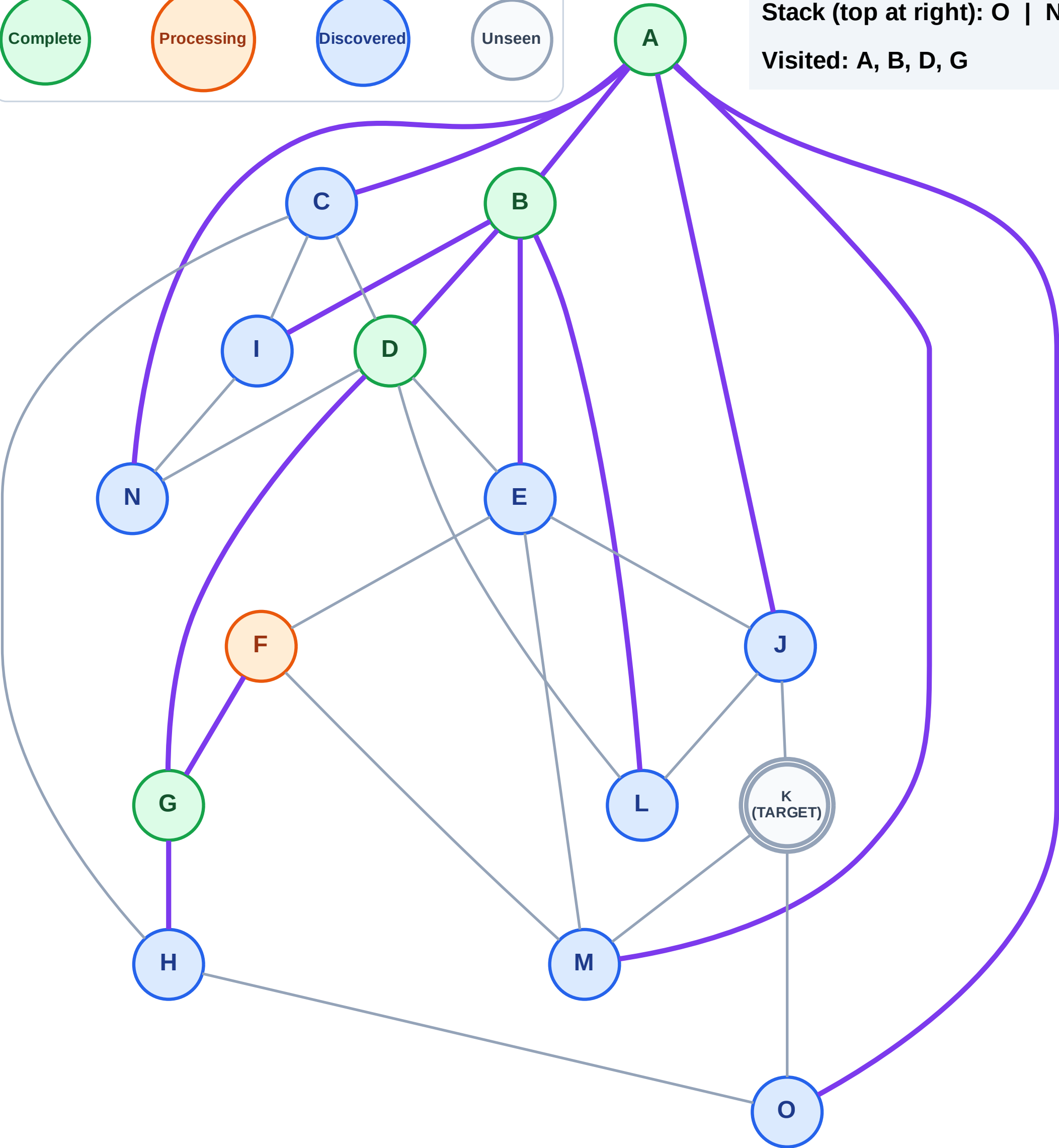
Processing

Discovered

Unseen

Stack (top at right): O | N | M | J | C | L | I | E | H

Visited: A, B, D, G



DFS Traversal

Step 11: No new neighbors from F

Legend

Complete

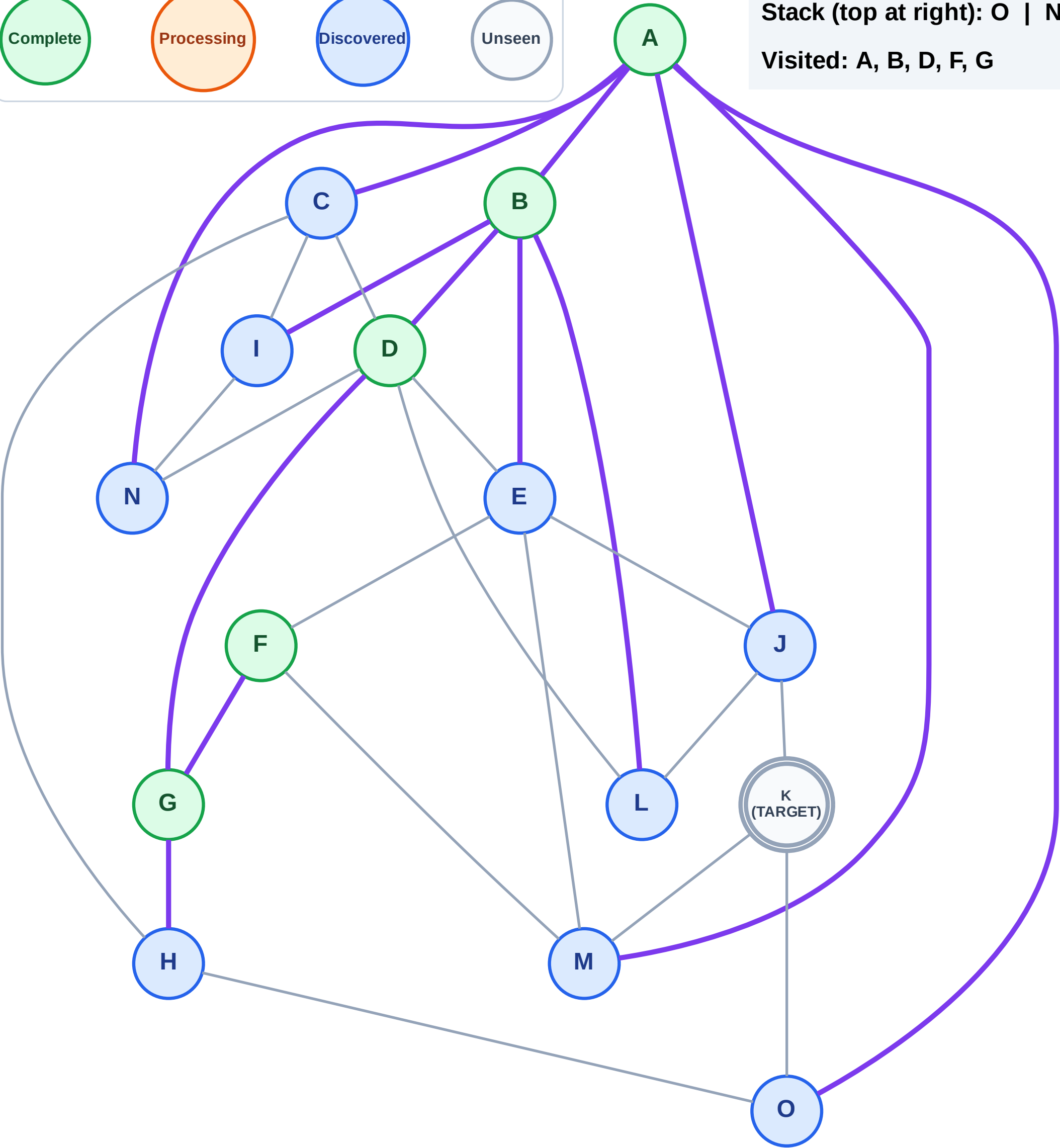
Processing

Discovered

Unseen

Stack (top at right): O | N | M | J | C | L | I | E | H

Visited: A, B, D, F, G



DFS Traversal

Step 12: Process node H

Legend

Complete

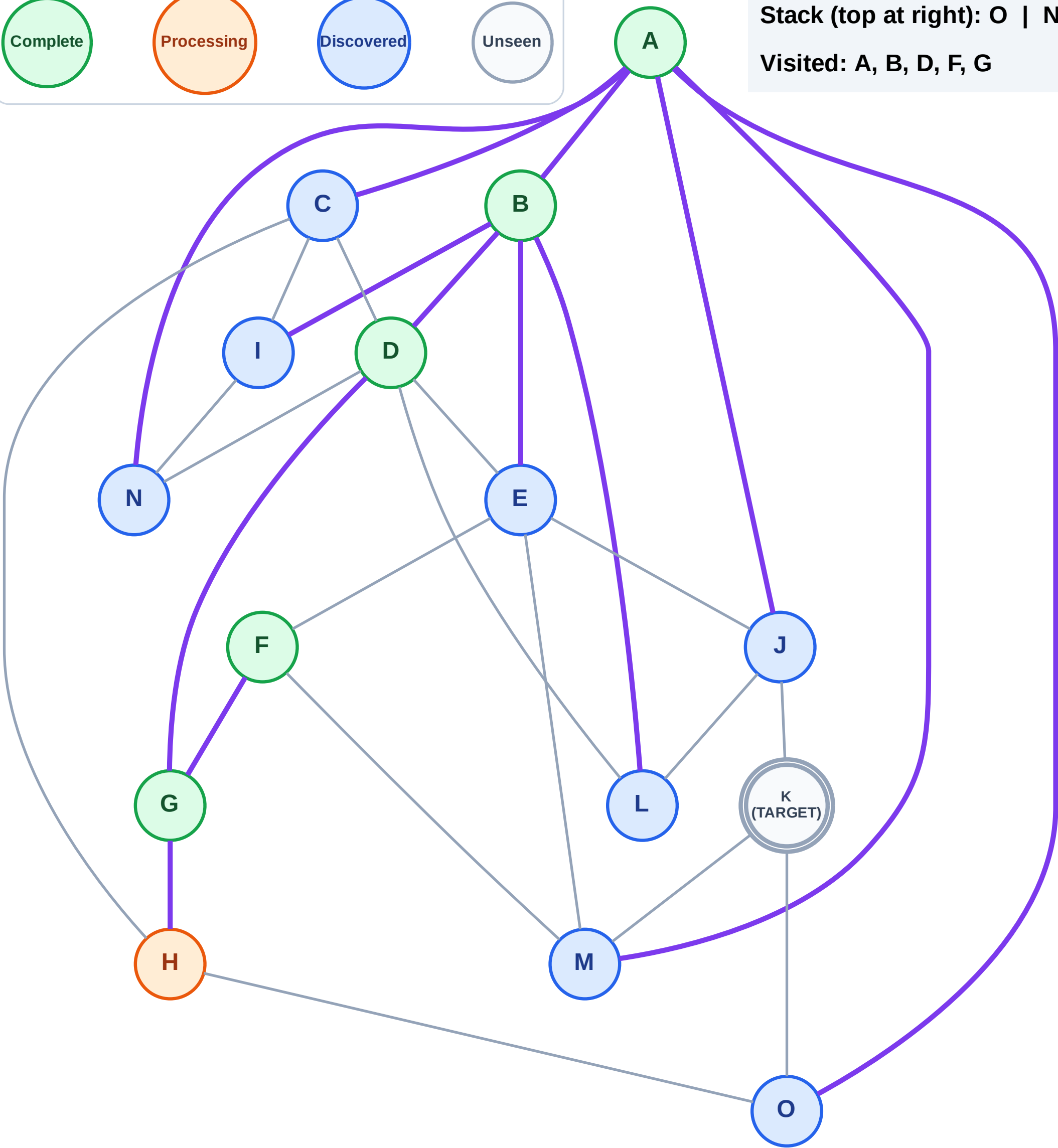
Processing

Discovered

Unseen

Stack (top at right): O | N | M | J | C | L | I | E

Visited: A, B, D, F, G



DFS Traversal

Step 13: No new neighbors from H

Legend

Complete

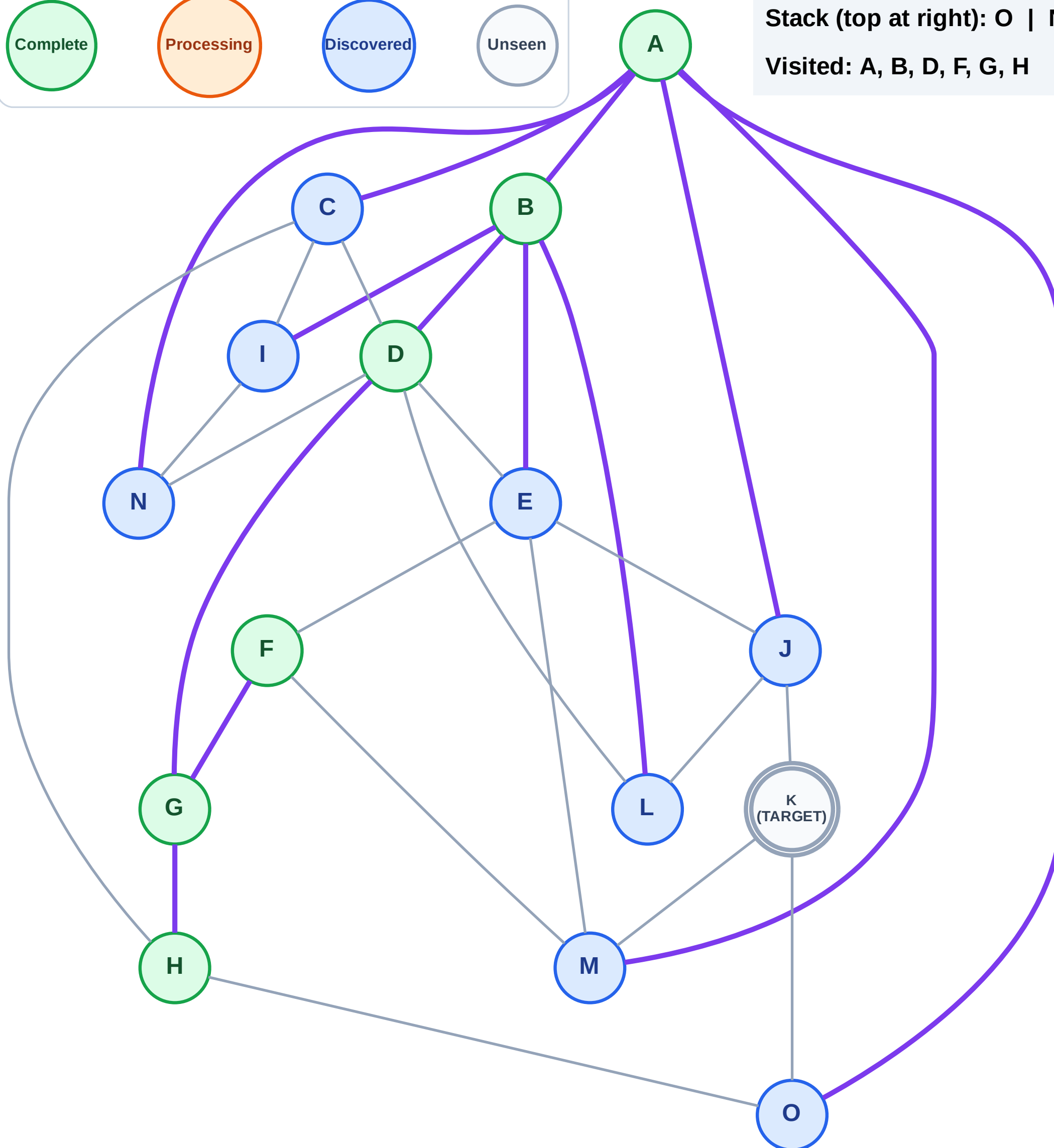
Processing

Discovered

Unseen

Stack (top at right): O | N | M | J | C | L | I | E

Visited: A, B, D, F, G, H



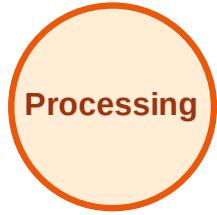
DFS Traversal

Step 14: Process node E

Legend



Complete



Processing



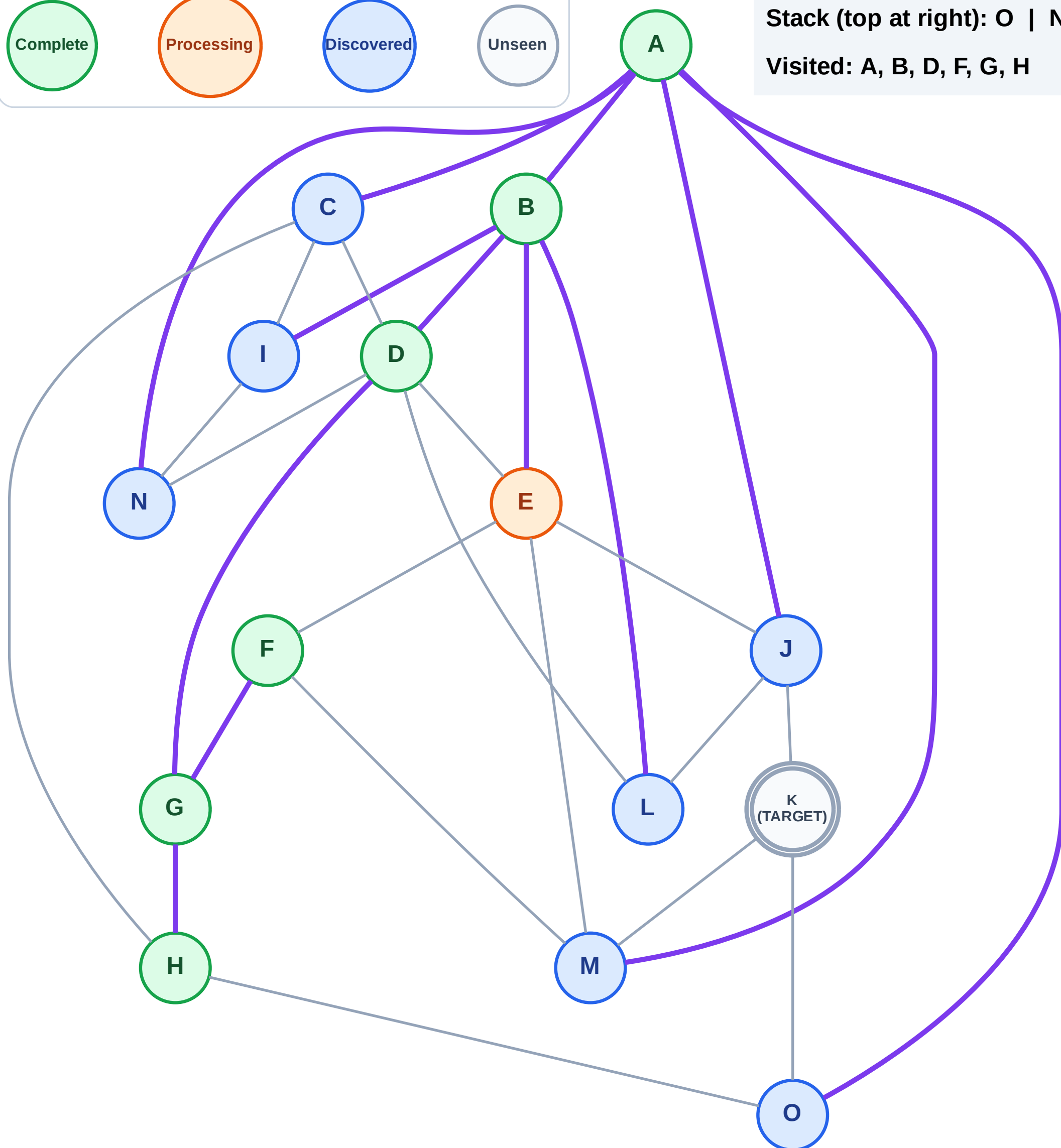
Discovered



Unseen

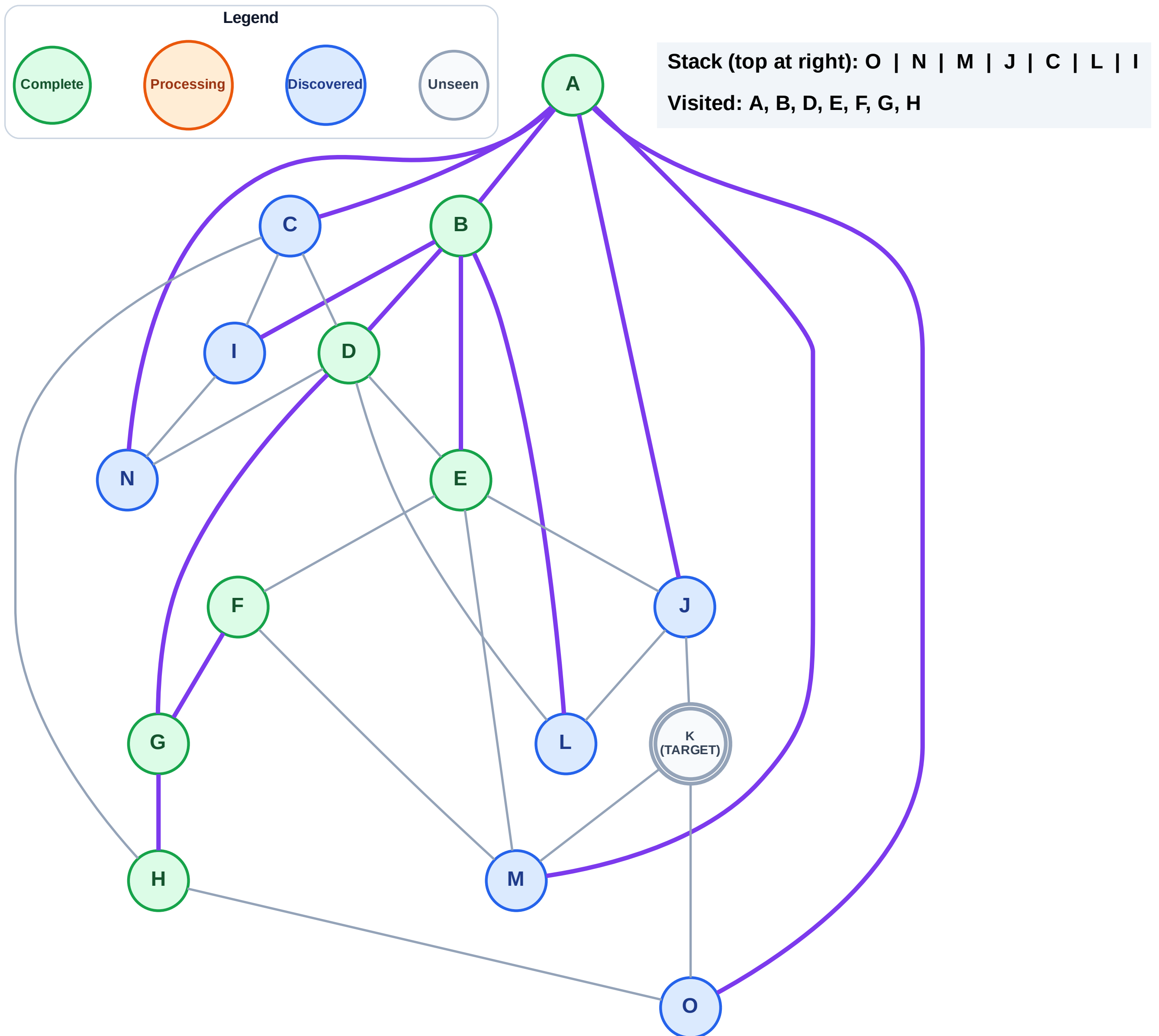
Stack (top at right): O | N | M | J | C | L | I

Visited: A, B, D, F, G, H



Step 15: No new neighbors from E

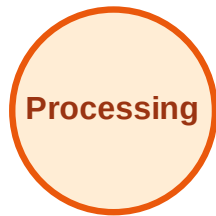
Step 15: No new neighbors from E



DFS Traversal

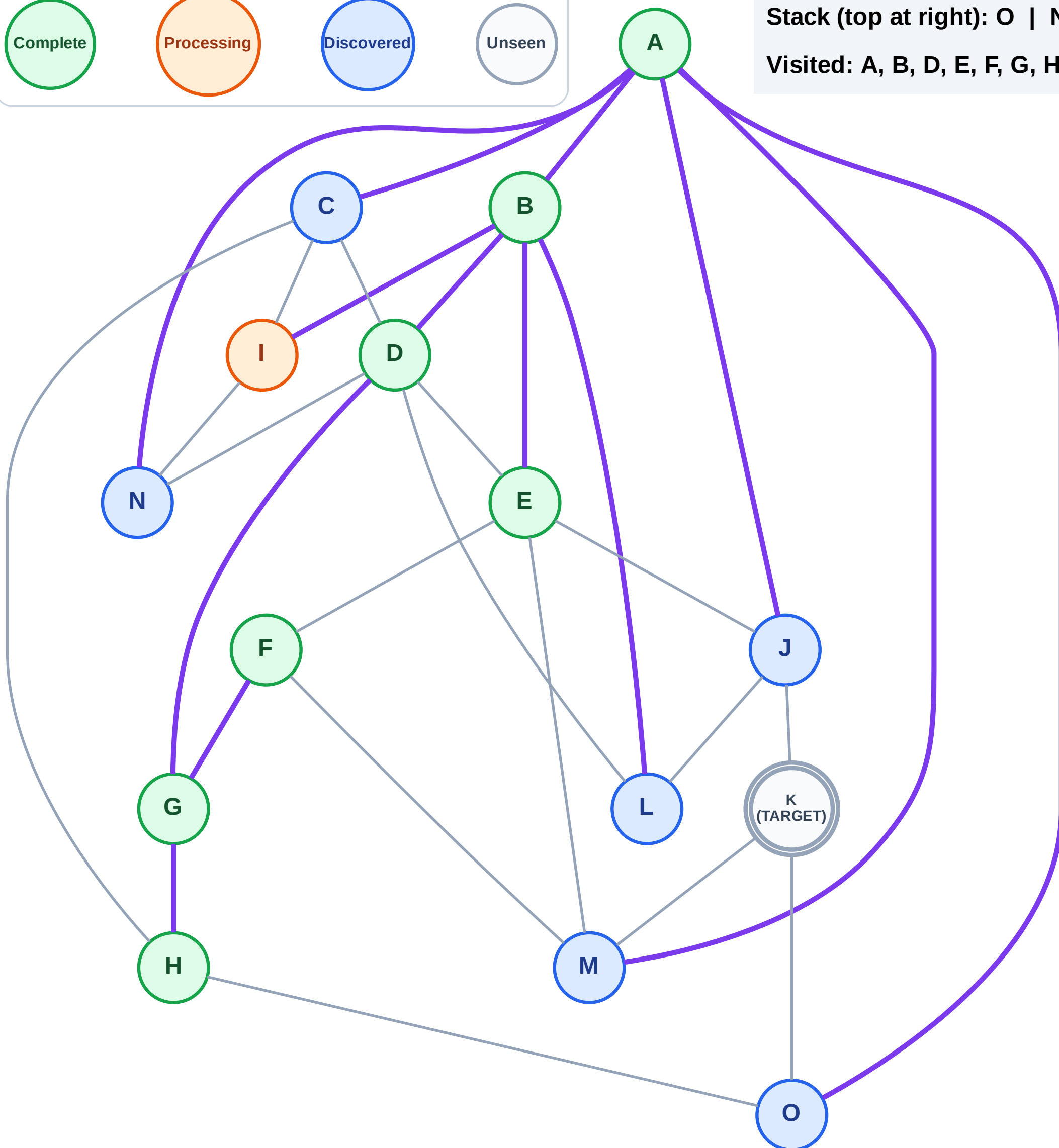
Step 16: Process node I

Legend



Stack (top at right): O | N | M | J | C | L

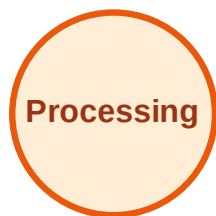
Visited: A, B, D, E, F, G, H



DFS Traversal

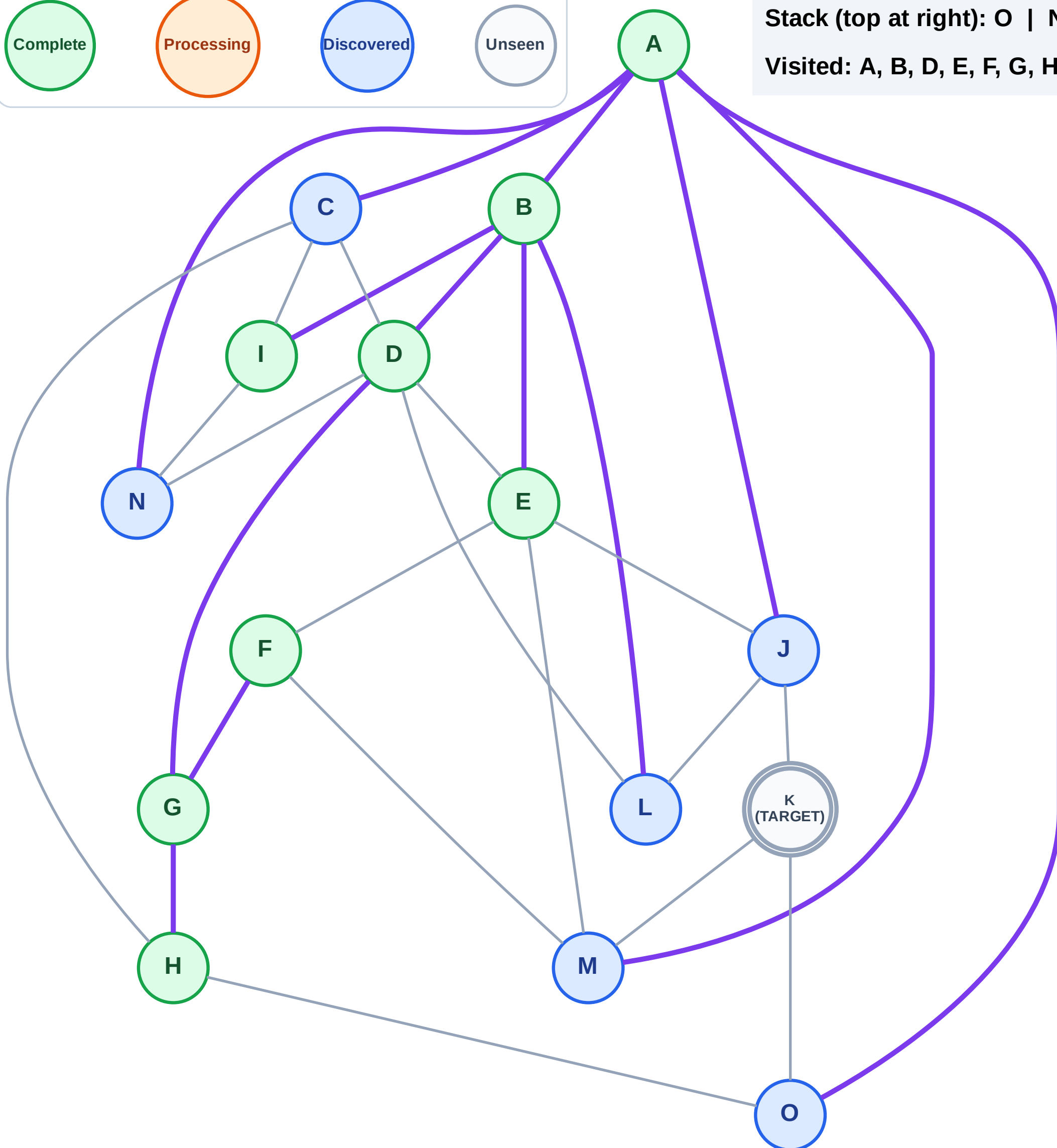
Step 17: No new neighbors from I

Legend

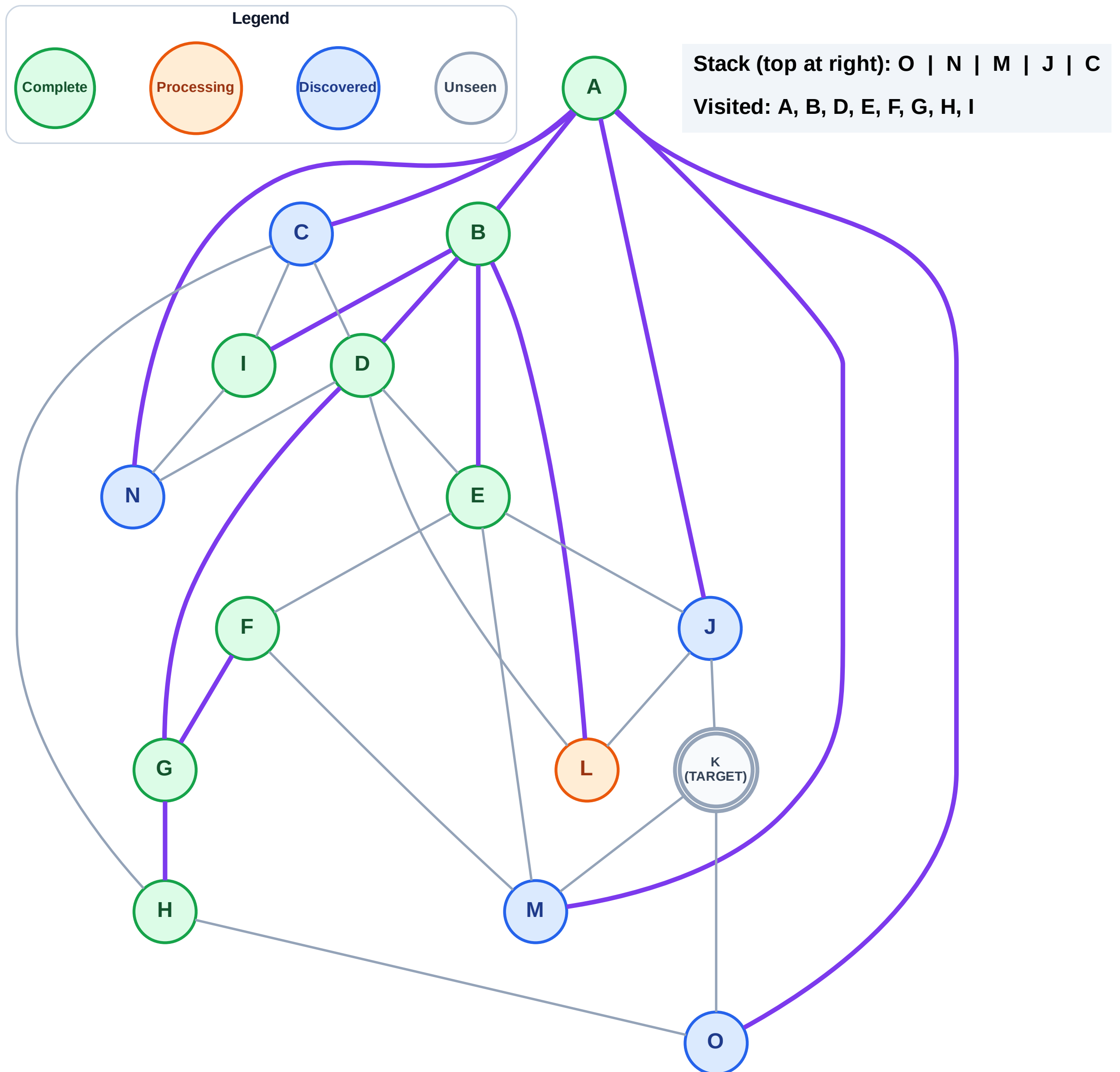


Stack (top at right): O | N | M | J | C | L

Visited: A, B, D, E, F, G, H, I



Step 18: Process node L



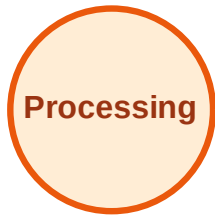
DFS Traversal

Step 19: No new neighbors from L

Legend



Complete



Processing



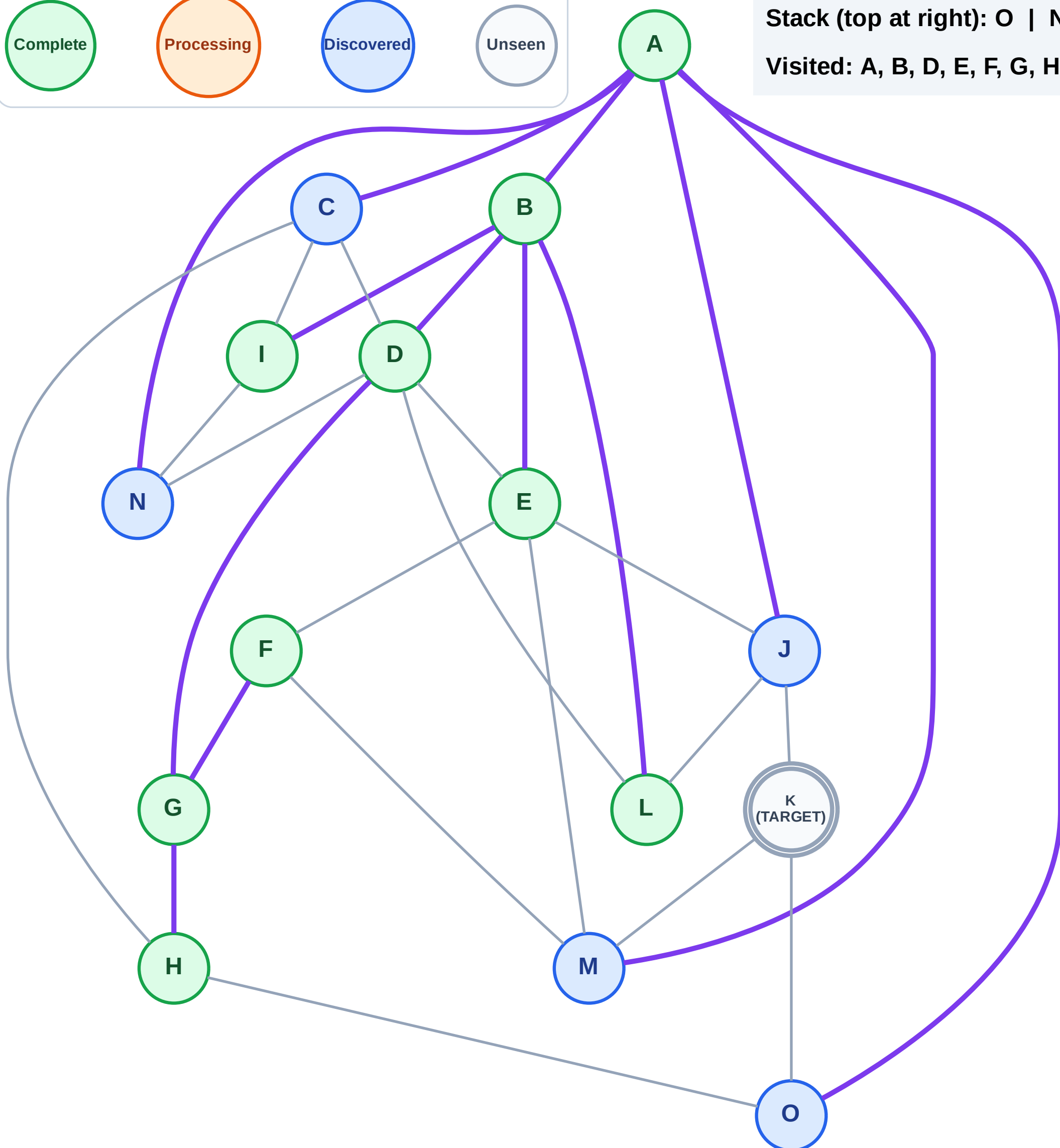
Discovered



Unseen

Stack (top at right): O | N | M | J | C

Visited: A, B, D, E, F, G, H, I, L



DFS Traversal

Step 20: Process node C

Legend

Complete

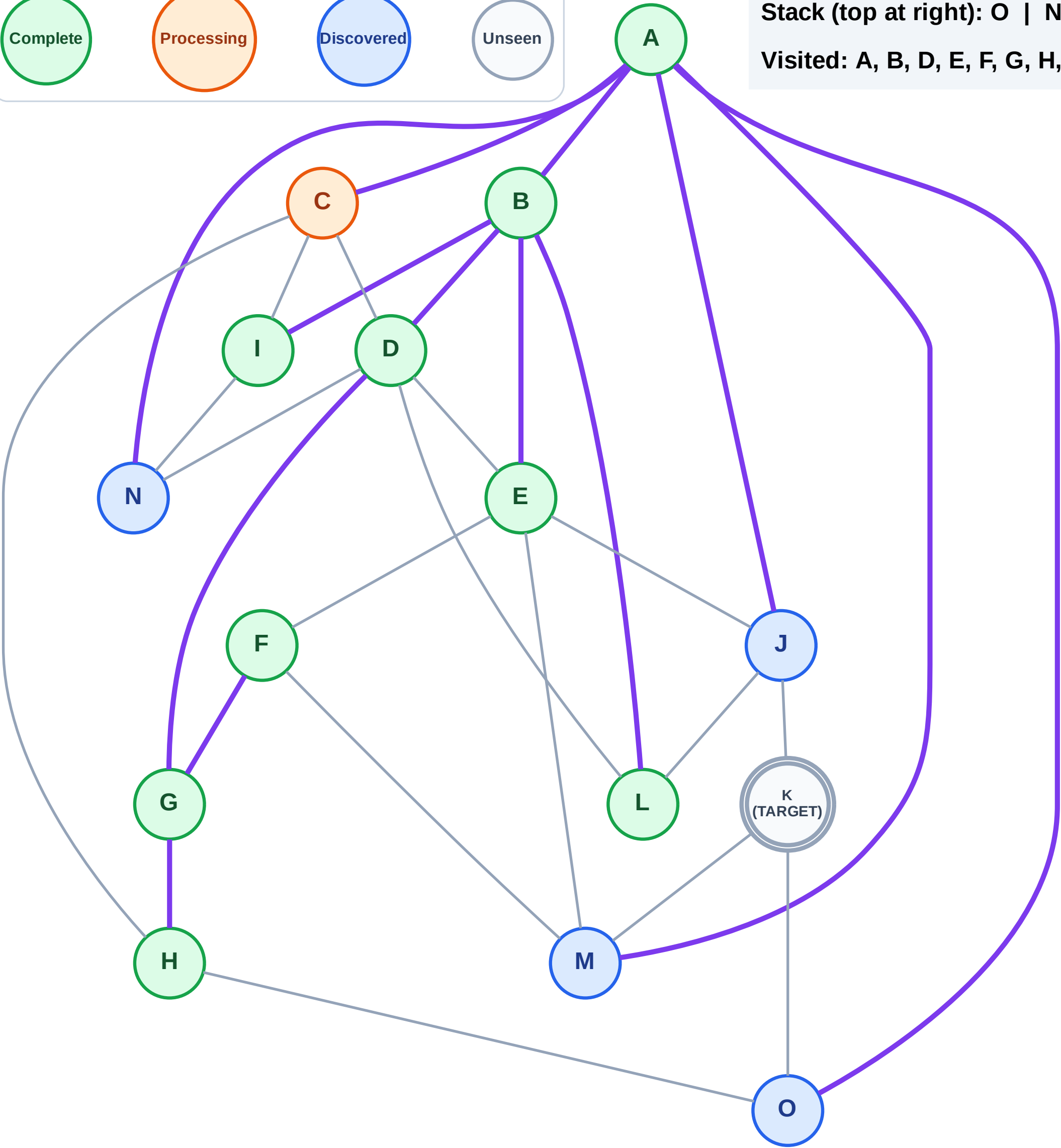
Processing

Discovered

Unseen

Stack (top at right): O | N | M | J

Visited: A, B, D, E, F, G, H, I, L



DFS Traversal

Step 21: No new neighbors from C

Legend

Complete

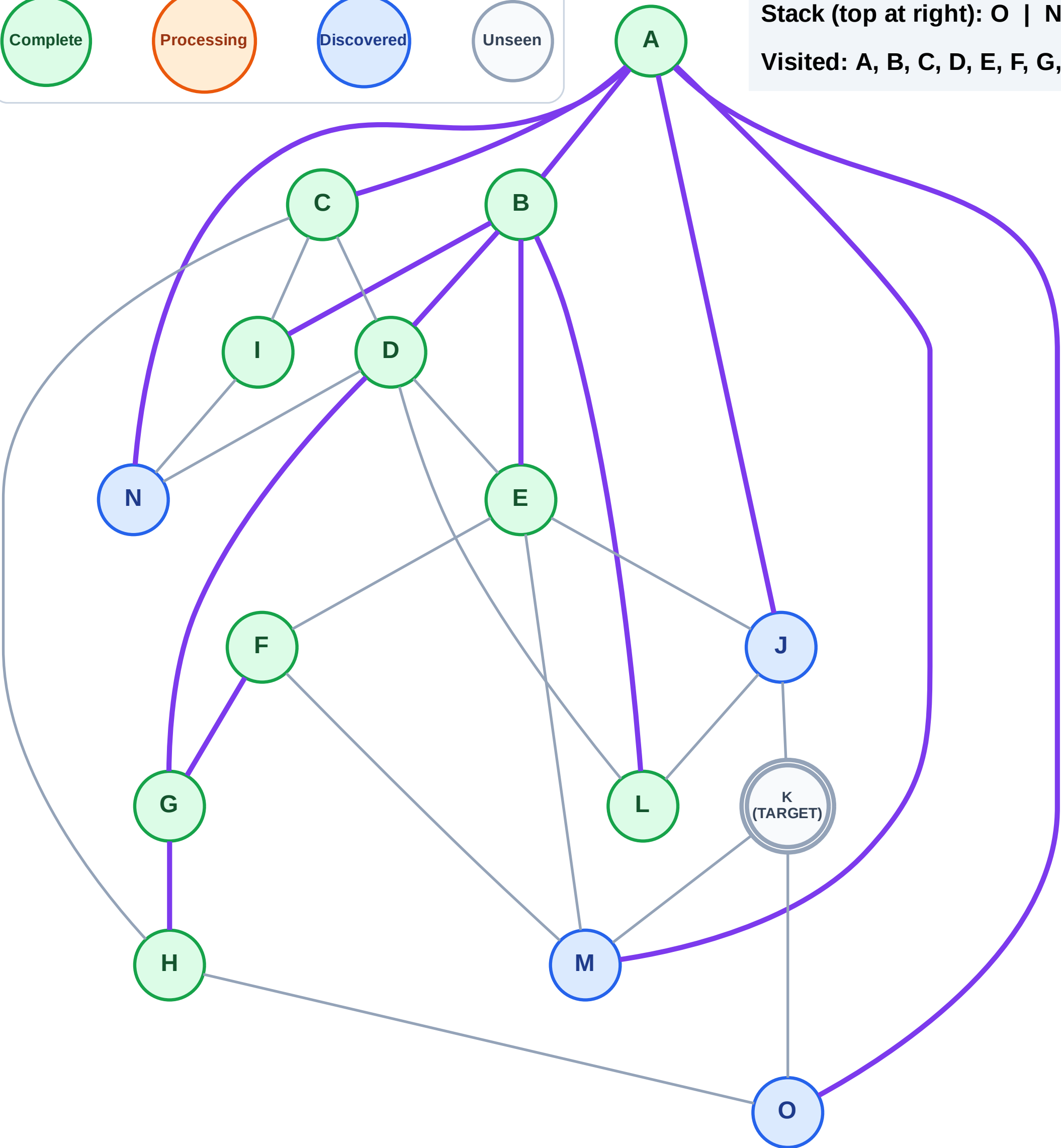
Processing

Discovered

Unseen

Stack (top at right): O | N | M | J

Visited: A, B, C, D, E, F, G, H, I, L



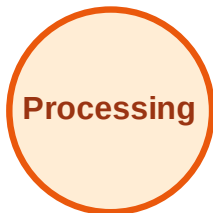
DFS Traversal

Step 22: Process node J

Legend



Complete



Processing



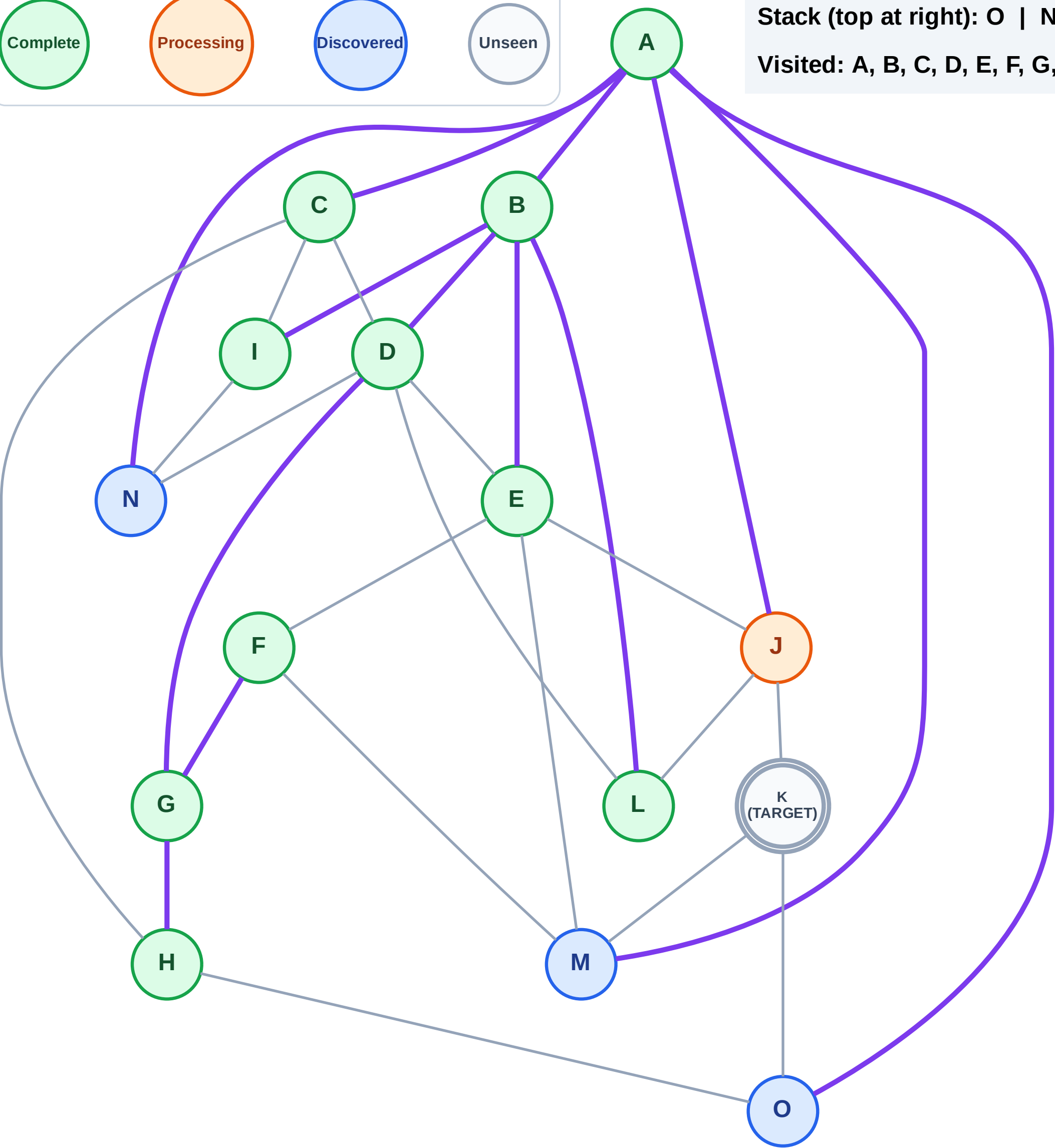
Discovered



Unseen

Stack (top at right): O | N | M

Visited: A, B, C, D, E, F, G, H, I, L



DFS Traversal

Step 23: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → J → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | N | M

Visited: A, B, C, D, E, F, G, H, I, J, K, L

